
When Does a Small Model Know to Hand Off? Zero-Training Escalation Gates for Full-Duplex Speech Models

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Abstract

1 Full-duplex speech models answer in real time with a small on-device backbone,
2 yet many queries exceed its competence. We ask whether such a model exposes
3 — with *zero additional training* — a pre-answer, modality-robust internal signal
4 reliable enough to hand a query off to a large cloud model mid-conversation, with-
5 out breaking the live session. On MiniCPM-o 4.5 we find the standard readouts
6 each fail in a systematic way: duplex fine-tuning re-purposes the late-layer last-
7 token position for turn control, breaking final-layer probes on text input, while
8 verbalized self-evaluation collapses on audio input. The information survives
9 underneath: a mid-network linear probe transfers from cheap text calibration to
10 speech, decides at end-of-turn in ~ 30 ms — before the first output token (AUC
11 .887) — and the same low-confidence state predicts that the talker will actually
12 relay the expert’s answer, so escalation and injection are one signal. Run live end-
13 to-end, the gated duplex system climbs monotonically from its honest streaming
14 floor (.422 $\rightarrow$.633 accuracy at 57% escalation), with the gap to offline operation
15 itemized into measured audio, streaming, and speech-channel taxes. The gate then
16 transfers *frozen* to six public speech benchmarks using only label-free threshold
17 quantiles — lifting Speech TriviaQA to .860 live against an official offline anchor
18 of .755 — and its mid-network readability replicates on an architecturally disjoint
19 second duplex family. Duplex models carry a free, readable competence signal
20 that standard readouts miss; read correctly, it turns a 9B talker into a system that
21 knows when to hand off.

22 1 Introduction

23 Voice assistants built on full-duplex speech models [Défossez et al., 2024, Zeng et al., 2024, Xu
24 et al., 2025] face a structural tension. The conversational loop—listening while speaking, turn-
25 taking inside a ~ 200 – 300 ms budget [Stivers et al., 2009]—demands a small, low-latency model
26 close to the user. Answer quality on knowledge- and reasoning-heavy queries demands a large
27 one. The natural architecture is a *talker-thinker* split, echoing the System 1/System 2 distinction
28 [Kahneman, 2011]: a small duplex “talker” holds the live session and answers by default, and a
29 large cloud “thinker” is consulted only when needed. This paper is about the load-bearing decision
30 in that architecture: **when should the talker hand off?**

31 We deliberately restrict ourselves to *zero-training* gate signals — quantities that can be read off the
32 small model as-is, without fine-tuning it: decode-time scalars (entropy, margin), linear probes on

hidden states [Alain and Bengio, 2016], and verbalized self-evaluation, i.e. $p(\text{True})$ -style “Would you answer this correctly?” prompts [Kadavath et al., 2022]. The constraint is practical (duplex fine-tunes are expensive and fragile to retrain) but it also sharpens the scientific question: *does a small full-duplex model already know what it does not know, and can that knowledge be read out in time to act on it?*

Our answer, developed over a controlled study of ten backbones/fine-tunes and both input modalities, is: **the knowledge is there; the readouts are what break, and they break in systematic, duplex-fine-tune-induced ways.** Concretely:

1. **Standard probes flatter to deceive (§4).** A final-layer hidden-state probe reaches AUC 0.822, but an audit shows it is mostly a query-*type* recognizer: a pool-identity oracle alone scores 0.715, and leave-one-pool-out (LOPO) transfer *inverts* on math (0.372). Pre-answer $p(\text{True})$ needs no training, beats the probe on most backbones, and scores 100%-fail SimpleQA traps at 0.945 — but only if asked *before* the model answers; after answering, it endorses its own confident-wrong answer (0.945 $\rightarrow$ 0.360).
2. **Duplex fine-tuning overwrites readouts, not knowledge (§5).** Across matched pairs (raw vs. omni-streaming vs. duplex fine-tunes of the same backbone), LOPO probe transfer degrades in the order raw > omni-streaming > duplex, with label coverage controlled. A full layer $\times$ position sweep localizes the damage: on MiniCPM-o 4.5, LOPO math is 0.931 at layer 22 but 0.366 at layer 35, with a sharp cliff over the last four layers, only at the last-token position — exactly where a streaming head must encode turn-control state.
3. **Each readout has one blind modality (§6).** The late-layer cliff is text-input-specific: under audio input the final layer does not invert (0.936). Conversely, verbalized introspection collapses on audio input (trap p_{yes} 0.055 $\rightarrow$ 0.556) on both MiniCPM duplex generations while omni-streaming and frozen-backbone controls stay intact. Three controlled arms (ASR audit, filler audio, audio+text duplication) pin the mechanism to *binding*: the verbal self-check runs over text-token pathways that audio embeddings do not feed, while a probe reads the same information at 0.93+.
4. **End-to-end training creates the shared core (§6).** Text $\leftrightarrow$ audio probe transfer is $\sim$ 0.86 on every end-to-end model but dead (0.52–0.60) on Freeze-Omni’s frozen backbone with identical text weights — so the practical recipe “calibrate on cheap text, deploy on speech” is valid for end-to-end models only.
5. **The gate wins as a system (§7).** The mid-layer probe decides in 20 ms (text) / 45 ms (audio), before the first output token. Against the model’s own `enable_thinking` mode, gated-cloud escalation dominates on both accuracy and latency at every escalation rate, because the gate flags knowledge/trap failures that thinking cannot fix — external escalation is *necessary*, not merely better.
6. **The loop closes live (§8).** In full streaming duplex sessions, an end-of-turn mid-layer read out-performs the offline reference (AUC .887 vs. .843) and still decides pre-first-token; the returned expert answer is spoken by the talker, and the pushback problem largely dissolves for *gated* escalations because the same low-confidence state that fires the gate predicts relay compliance (comply .42 $\rightarrow$.75 across gate-score quartiles). The live system climbs monotonically from its honest streaming floor (.422 $\rightarrow$.633 at 57% escalation on speakable content), with the gap to offline operation itemized into measured audio, streaming, and speech-channel taxes.
7. **The gate transfers frozen (§9).** With probe weights fixed and only label-free threshold quantiles per pool, the same gate runs the live loop on six public speech benchmarks — lifting Speech TriviaQA to .860 at 50% escalation against the model’s official offline .755, matching the cloud expert at half the expert calls on Llama Questions — and the mid-network readability it exploits replicates on a second, architecturally disjoint duplex family (hybrid-Mamba NemotronLabs-VoiceChat-11B, §5.4).

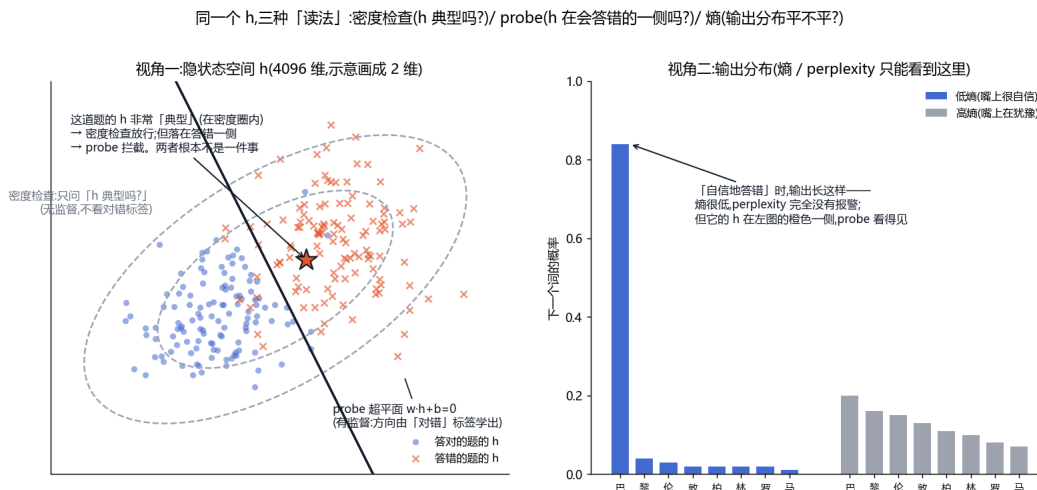


Figure 1: Conceptual contrast between a supervised probe in hidden-state space and what output-distribution scalars (entropy/perplexity) can see. **[TODO: regenerate this figure with English labels (current version has Chinese axis/legend text); source: figures/concept_probe_vs_ppl.py]**

The paper executes **both steps of the two-step design**. Step 1 (§4–§7) establishes when the talker should hand off; step 2 (§8) returns the thinker’s result into the live duplex session. Its anticipated obstacle — the talker pushing back on injected results that conflict with its own reasoning — turns out to be governed by the gate’s own signal: escalations the gate fires are exactly the ones the talker is willing to relay, and where confidence is high, authority *framing* works while forcing words into the talker’s mouth backfires (§8.2).

2 Related Work

Full-duplex real-time dialogue. End-to-end spoken dialogue models now span native duplex designs — dGSLM [Nguyen et al., 2023], Moshi with its Inner Monologue [Défossez et al., 2024], GLM-4-Voice [Zeng et al., 2024], and single-LLM native full-duplex [Fang et al., 2026] — streaming omni models such as Qwen2.5-Omni [Xu et al., 2025] and our target MiniCPM-o 4.5 [Yao et al., 2024, Cui et al., 2026], a 9B omni/duplex fine-tune of Qwen3-8B [Yang et al., 2025], alongside frozen-backbone adapters (Freeze-Omni, Wang et al., 2024a) and audio-understanding fine-tunes [Chu et al., 2024]. A parallel line adds explicit *turn-taking control*: FlexDuo [Liao et al., 2025] drives a Speak/Listen/Idle FSM at 120 ms ticks, MinMo [Chen et al., 2025] and Freeze-Omni attach lightweight state heads over the LLM’s own hidden states, and recent work analyzes synchronization and turn-taking behavior in duplex models directly [Riera et al., 2026]; Wu et al. [2026] interleave hidden reasoning with speech inside the duplex loop. All of these decide *when the local model should speak*; our gate decides *whether the local model should be the one answering at all*. The two decisions are orthogonal and mechanically compatible — the head-based controllers read the same hidden states our gate does, and their arbitration cadences (120–250 ms) show a live duplex loop already tolerates far more decision latency than our ~30 ms end-of-turn read. Their shared choice of a *last-layer* readout is one point on our per-layer curve — the point our results show duplex fine-tuning specifically damages (§5). We use this model landscape as a controlled independent variable: matched pairs across raw backbones, audio-only, streaming-omni, duplex, and adapter designs attribute the readout damage to duplex-ness specifically.

Does the model know what it knows? Kadavath et al. [2022] introduced $p(\text{True})$ self-evaluation; Lin et al. [2022] trained verbalized uncertainty. A parallel line reads correctness from internal states with linear probes [Alain and Bengio, 2016, Azaria and Mitchell, 2023, Marks and Tegmark, 2024,

Burns et al., 2023], and intermediate layers carry more truthfulness information than the output distribution reveals [Orgad et al., 2025] — consonant with our mid-layer findings. Semantic entropy [Kuhn et al., 2023, Farquhar et al., 2024] needs multiple sampled generations, which a real-time loop cannot afford; Semantic Entropy Probes [Kossen et al., 2024] amortize it into a cheap hidden-state probe — the closest signal to ours, but read on text-only, half-duplex models after the fact, not at end-of-turn inside a streaming session. Multimodal uncertainty estimation [Zhang et al., 2026] asks whether confidence can be fused across modalities; we ask the prior question of *where* in a duplex model a modality-robust confidence read survives at all. Our contribution to this literature is not a new signal but a controlled account of when the standard readouts break: probe and $p(\text{True})$ readouts fail in complementary, modality- and architecture-specific ways under duplex fine-tuning, while the underlying information survives mid-network.

Selective escalation and routing. Cost-quality routing across LLMs is well studied: FrugalGPT [Chen et al., 2023] cascades with a learned scorer, HybridLLM [Ding et al., 2024] and RouteLLM [Ong et al., 2024] train query routers, RouterBench [Hu et al., 2024] standardizes evaluation, and Li et al. [2026] report that many routers fail to beat simple baselines under a unified protocol. AutoMix [Aggarwal et al., 2024] is the closest system design — a small model decides its own answer’s reliability and escalates — but the decision is *post-answer* self-verification on text; our gate fires *pre-answer*, before the first output token, which is what a voice deadline requires. Confidence-based cascades also appear at the serving layer [Xue et al., 2025]. These routers decide from query-side features, and the large model then answers the user directly; both assumptions fail in full-duplex voice, where the user talks to one voice whose session state lives in the small model. Our audit (§4) quantifies the query-side ceiling: it is the “type shortcut” whose transfer collapses under distribution shift, while the internal mid-layer signal transfers. Learning-to-defer [Madras et al., 2018] and early exit [Schuster et al., 2022] make the same decision inside one model; speculative decoding [Leviathan et al., 2023] inverts our roles, and our prefetch analysis (§8.1) is its query-level mirror. Escalating to the model’s own thinking mode connects to test-time compute [DeepSeek-AI et al., 2025]; we find it dominated by external escalation because the flagged failures are knowledge-bound, not compute-bound.

The gap. Duplex work decides *when to speak*; routing work decides *which model* from the query surface or after a first answer; uncertainty work asks *whether the model is confident*, almost always in text and offline. To our knowledge, no prior work asks whether a full-duplex speech model exposes a *pre-answer*, *modality-robust*, *internal* competence signal early enough to drive a live small-to-large handoff while preserving the ongoing conversational session — the regime this paper measures end-to-end, from signal (§4–5) to live system (§8) to frozen transfer (§9).

Evaluation practice. Query pools come from public benchmarks only: SimpleQA [Wei et al., 2024] for confidently-wrong long-tail traps (cf. PopQA, Mullen et al., 2023), MMLU-Pro [Wang et al., 2024b], TriviaQA [Joshi et al., 2017], GSM8K [Cobbe et al., 2021], MATH-500 [Hendrycks et al., 2021, Lightman et al., 2024], and Dolly/Alpaca instruction data [Conover et al., 2023, Taori et al., 2023]; adequacy labels come from an LLM judge [Zheng et al., 2023] validated by a stronger re-judge (agreement 0.962 text / 0.945 audio). The audio arm TTS-renders the frozen pool, following Spoken-SQuAD [Li et al., 2018] and VoiceBench [Chen et al., 2024], and the audio-side findings are validated on *real* human speech via SD-QA [Faisal et al., 2021] (§6) and the live external benchmarks of §9.

3 Experimental Setup

Target model and matched-pair controls. The system target is **MiniCPM-o 4.5** (9B; omni + full-duplex fine-tune of Qwen3-8B), run in bf16 with greedy decoding, seed 42. To attribute effects to the *kind* of fine-tune rather than the backbone, we assemble matched pairs and controls (Table 1): the raw backbone family member for each fine-tune, a streaming-omni fine-tune (Qwen2.5-Omni-7B), a second-generation duplex model (MiniCPM-o 2.6), an audio-understanding fine-tune (Qwen2-

Table 1: Model roster. “Pair” groups a fine-tune with its raw backbone control; the comparison statistic is always the within-pair difference.

model	type	pair	role
MiniCPM-o 4.5	omni + duplex FT	Qwen3-8B	system target
Qwen3-8B	raw	—	pair control
MiniCPM-o 2.6	duplex FT	Qwen2.5-7B	duplex replication
Qwen2.5-Omni-7B	streaming omni FT	Qwen2.5-7B	omni control
Qwen2.5-7B	raw	—	pair control
Qwen2-Audio-7B	audio FT	Qwen2-7B	audio-FT control
Freeze-Omni	frozen LLM + adapter	Qwen2-7B	adapter control
Qwen2-7B	raw	—	pair control
Mistral-7B-v0.3	raw	—	backbone diversity
GLM4-9B-chat	raw	—	backbone diversity

Audio-7B), a frozen-backbone adapter model (Freeze-Omni, whose LLM weights are byte-identical to its raw control), and two additional raw backbones for diversity (Mistral-7B-v0.3 [Jiang et al., 2023], GLM4-9B-chat [Team GLM et al., 2024]); the Qwen2.5/Qwen2 raw controls follow Yang et al. [2024].

Query pool (frozen, public benchmarks only). 600 queries in five pools, frozen before any gate experiment (360 calibration / 240 test, seed 42): *trap* = SimpleQA (50; the target model fails 100%), *hard-knowledge* = MMLU-Pro (150), *easy-fact* = TriviaQA (100), *easy-chat* = Dolly-15k (en) + Alpaca-zh (150), *hard-math* = GSM8K tail + MATH-500 (150). Text-input fail rates for MiniCPM-o 4.5: *trap* 1.00 / *knowledge* .48 / *fact* .34 / *chat* .21 / *math* .19 (overall .358). We deliberately use no LLM-generated queries: a pilot GPT-generated *trap* pool failed its design (90% answered correctly by the 9B target) and creates a generator-equals-judge circularity.

Gate signals (all zero-training). (i) *Decode scalars*: max token entropy / margin over the first k decode steps. (ii) *Linear probe*: logistic regression on a single hidden state, position $\times$ layer swept; trained on calibration features, thresholds from out-of-fold scores. (iii) *Verbalized self-evaluation* [Kadavath et al., 2022]: $p(\text{True})$ -pre reads $P(\text{Yes})$ off the first token of “Would you answer this correctly?” asked *before* answering; $p(\text{True})$ -post asks “Is this answer correct?” about a drafted answer.

Labels and judge. Adequacy labels come from gpt-5.4-mini with structured outputs, blind to the gate. The judge is validated by a gpt-5.5 re-judge: agreement 0.962 (text) / 0.945 (audio); every headline number moves ≤ 0.010 under the stronger judge’s labels. Escalation target for system experiments: gpt-5.5. [TODO: cite/describe judge models properly; OpenAI model documentation entry in refs.bib is a placeholder]

Audio arm. The frozen pool is TTS-rendered (OpenAI tts-1, single voice), so query *content* is unchanged and only modality varies, following Spoken-SQuAD / VoiceBench practice [Li et al., 2018, Chen et al., 2024]. Real-speech validation on SD-QA [Faisal et al., 2021] is open (§11).

Infrastructure. All experiments run on Modal H100 instances; torch 2.8.0, transformers 4.51.0 (4.57.6 for Qwen2.5-Omni; 4.45.2 for Freeze-Omni). Total compute + API spend for Phases 0–7a: $\approx \$290$.

4 Which Signal Knows the Model Will Fail? (Text Input)

4.1 Headline comparison

Table 2 compares the candidate signals on MiniCPM-o 4.5 (calibration split; probe numbers are 5-fold out-of-fold). Figure 2 shows the ROC curves.

Table 2: Signal AUCs against judged failure labels, MiniCPM-o 4.5, text input. Bracketed interval is a bootstrap 95% CI.

signal	AUC	training needed
$p(\text{True})$ -post (post-draft)	0.899	none
probe on h_{prompt} (final layer)	0.822 [0.777, 0.863]	LR on calib
$p(\text{True})$ -pre (pre-answer)	0.807	none
max entropy@4 (best decode scalar)	0.696	none

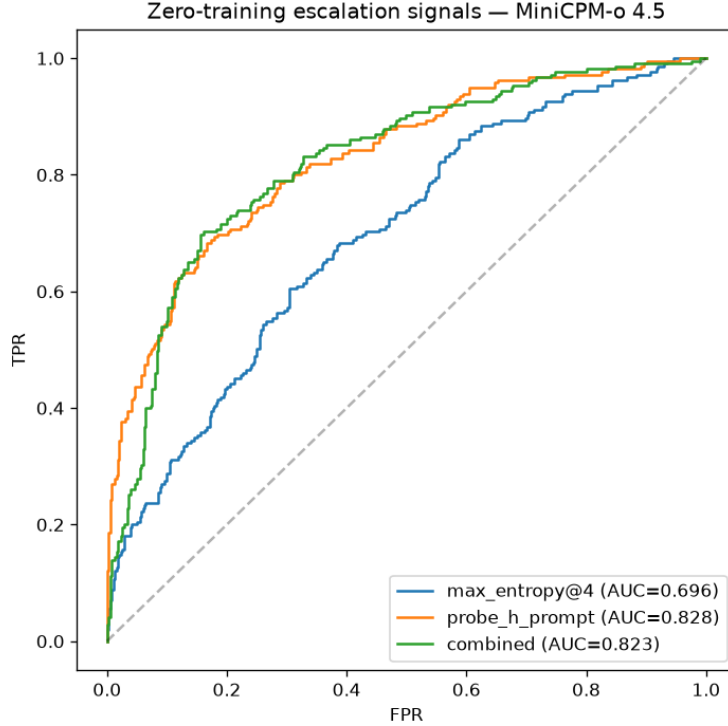


Figure 2: ROC of the final-layer probe vs. decode scalars (calibration split). **[TODO: regenerate to include the $p(\text{True})$ curves on the same axes]**

193 4.2 The probe’s number is a type shortcut

194 The probe’s 0.822 does not mean what it appears to mean. Three audit results (all calibration-
 195 only): (i) the same hidden state classifies the *source pool* of a query at 95.8%; (ii) a pool-identity
 196 oracle alone — predict each pool’s base fail rate — already reaches AUC 0.715; (iii) appending true
 197 pool labels to the probe adds nothing ($0.822 \rightarrow 0.821$), i.e. the probe has already absorbed the type
 198 information. The decisive test is leave-one-pool-out (LOPO) transfer: held-out math *inverts* to 0.372
 199 — the probe scores terse MATH-500 items as “safer” than verbose GSM8K items, a style-familiarity
 200 artifact — and the mean escalation score on the 100%-fail trap pool is 0.232. Decode entropy is
 201 $\approx$ chance on two of three raw backbones tested. In-distribution AUC therefore overstates what a
 202 query-side signal knows; we treat LOPO as the primary metric for signal quality throughout.

203 4.3 $p(\text{True})$: ask before the model answers

204 Verbalized self-evaluation needs no training and transfers differently. On the 100%-fail SimpleQA
 205 trap pool, pre-answer $p(\text{True})$ says “No, I will not get this right” with mean escalation score 0.945
 206 — on the same items the probe scores 0.232. The timing is essential: asked *after* the model has
 207 produced a confident-wrong answer, self-assessment collapses to 0.360 (the model endorses its own

Table 3: LOPO transfer (AUC on the held-out pool) of the final-layer last-token probe, Qwen2.5-7B family, label-coverage matched. The o4.5 column is the Qwen3-family pair for reference.

LOPO pool	Qwen2.5-7B (raw)	Qwen2.5-Omni	MiniCPM-o 2.6 (duplex)	[o4.5 vs Qwen3]
hard-math	0.809	0.745	0.538 $\approx$ chance	0.372 (inverts)
hard-knowledge	0.680	0.613	0.526 $\approx$ chance	0.61

Table 4: LOPO math transfer by depth (last-token probes).

model	best mid-layer	final layer	shape
Qwen3-8B (raw)	0.964 (L33/36)	0.958	plateau, no cliff
MiniCPM-o 4.5 (duplex)	0.931 (L22/36)	0.366	cliff L31 $\rightarrow$ L34, inverts
Qwen2.5-7B (raw)	0.893 (L21/28)	0.809	mild dip
Qwen2.5-Omni (streaming)	0.794 (L16/28)	0.746	diffuse mild depression
MiniCPM-o 2.6 (duplex)	0.822 (L21/28)	0.540	cliff L21 $\rightarrow$ L27

answer). Post-draft $p(\text{True})$ (“Is this answer correct?”) is the strongest overall signal (0.899) but requires paying for a full draft first. Across backbones, $p(\text{True})$ -post beats the trained probe on four of five raw/duplex models tested; the exception (GLM4-9B-chat, 0.685 vs. probe 0.798) shows the pattern depends on the backbone’s self-evaluation calibration quality rather than being universal.

5 Duplex Fine-Tuning Overwrites Readouts, Not Knowledge

5.1 Matched pairs: damage tracks duplex-ness, not audio

Is the LOPO transfer failure of §4.2 a MiniCPM quirk? Table 3 shows LOPO transfer of the standard (final-layer, last-token) probe across one backbone family, with label coverage matched by subsampling the raw control to the duplex model’s per-pool fail rates (`lopo_matched`): the raw backbone matched to MiniCPM-o 2.6’s exact fail rates keeps math at 0.825 while the duplex model itself sits at chance — so label coverage is ruled out and the *representation* changed. The degradation is ordered: raw > omni-streaming > duplex. Audio-only fine-tuning is separately controlled (Qwen2-Audio), though that pair is confounded by capability collapse and is reported as a documented negative. Raw-backbone baselines vary across families, so the defensible statistic is always the within-pair difference.

5.2 Layer $\times$ position sweep: the cliff

We capture every decoder layer at prefill, at both the last-token and mean-pooled positions (final-layer values reproduce the generation-time hooks exactly, 5-for-5). Figure 3 and Table 4 show the central result: **mid-network, the duplex model carries near-raw transferable difficulty information; the damage is confined to the last few layers at the last-token position.**

Three structural facts localize the mechanism. First, the within-pair deficit at the best mid-layer is only -0.03 , vs. -0.59 at the final layer. Second, mean-pooled probes survive to the final layer on duplex models — the damage is (late-layer $\times$ last-token)-specific, exactly the position a streaming/duplex head must re-purpose to encode turn-control state. Third, the two fine-tune styles fail differently: duplex damage is severe but *local*; omni-streaming damage is mild but *diffuse*. We conclude the duplex fine-tune **overwrites the readout** rather than destroying the model’s self-knowledge — which reverses the natural interpretation of §4.2: the signal exists and is strong; the standard readout simply cannot see it on duplex models.

5.3 Does the mid-layer convert into a better gate?

On the frozen test split, accuracy-vs-escalation-rate areas over the random baseline order as: final-layer probe $+0.054 < p(\text{True})\text{-pre} +0.059 < \text{mid-layer L22 probe} +0.064 < p(\text{True})\text{-post}$

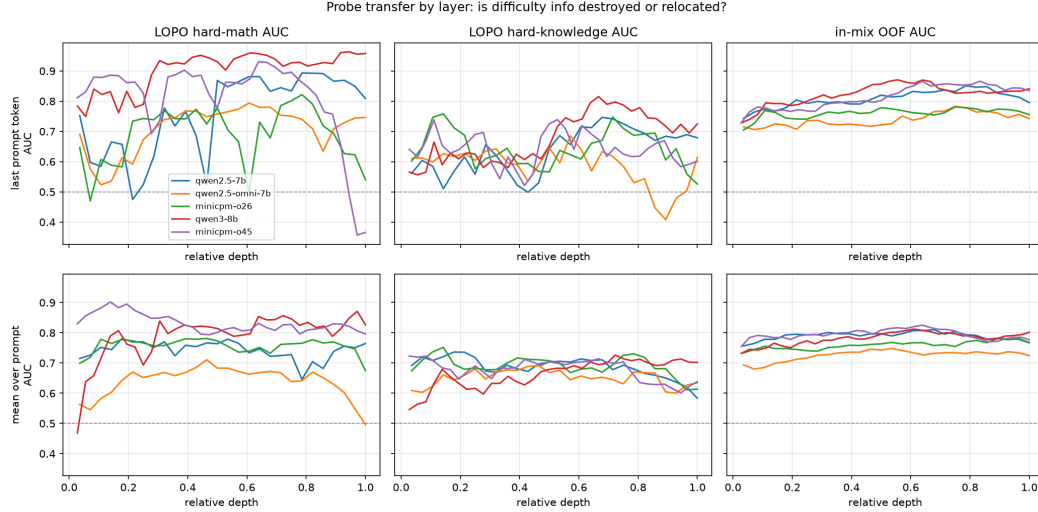


Figure 3: LOPO transfer by layer and position. Duplex models (MiniCPM-o 4.5/2.6) show a late-layer cliff at the last-token position only; mean-pooled probes survive to the final layer; raw backbones show no cliff; the omni-streaming control shows mild diffuse depression instead.

239 +0.068 (which requires a full draft). The optimum is not knife-edge (L20–L30 all $\geq +0.056$; mean-
 240 pool uniformly worse), and the in-mix area *under-sells* the mid-layer probe: its real advantage is
 241 LOPO robustness (math 0.93 vs. 0.37), which a same-mix test cannot show (Figure 4). Caveat:
 242 quantile-threshold transfer of probe scores remains weak and needs deployment-grade calibration
 243 (§11); AUC/area claims are unaffected.

244 5.4 A second duplex family: the pre-registered test executed

245 The mid-band claim was pre-registered as a prediction: when another open-weight full-duplex fam-
 246 ily became available, its usable read should also sit mid-network, not at the standard (final-layer,
 247 last-token) readout. NVIDIA NemotronLabs-VoiceChat-11B — a hybrid Mamba/attention back-
 248 bone (Nemotron Nano v2 9B: 56 blocks, 27 Mamba-2 / 4 attention), i.e. architecturally as far from
 249 MiniCPM’s dense transformer as the open-weight space currently offers — provides the test, and
 250 it passes on first execution. With the identical recipe (same wavs, same judge, 600-query cali-
 251 bration only), the layer sweep peaks mid-band at L30–34 (eot-last AUC .714 vs. .693/.682 at the
 252 endpoints); stacking the same three zero-latency positions lifts OOF .714 $\rightarrow$.761 $\rightarrow$ **.790**; and
 253 the frozen-methodology probe transfers to the external pools at AUC .781/.793/.754/.701 (strivi-
 254 aqa/swebq/sdqa/sllama) — the MiniCPM v3 band reached with a quarter of the calibration data
 255 (Figure 5). Two boundary notes: the Nemotron backbone is substantially weaker at knowledge
 256 retrieval (striviaqa local accuracy .32 vs. MiniCPM’s .62, same judge), and it is English-only —
 257 Chinese audio yields fluent unrelated English, so the cross-lingual pool has no analog here (zero la-
 258 bel variance; skipped). The live 4-arm port and the trained-checkpoint ablation remain future work;
 259 the replicated claim is the structural one: *mid-network readability of the self-assessment signal is a*
 260 *property of duplex models as a class, not of one checkpoint.*

261 6 Speech Input: Each Readout Has One Blind Modality

262 6.1 The late-layer cliff is text-input-specific

263 Running the identical layer sweep on the TTS-rendered pool (audio input, content unchanged):
 264 audio $\rightarrow$ audio LOPO math reaches 0.93–0.96 at mid-layers *and the final layer does not invert* (L35
 265 = 0.936, vs. 0.366 on text). A speak-mode template control (use_tts_template=True on text

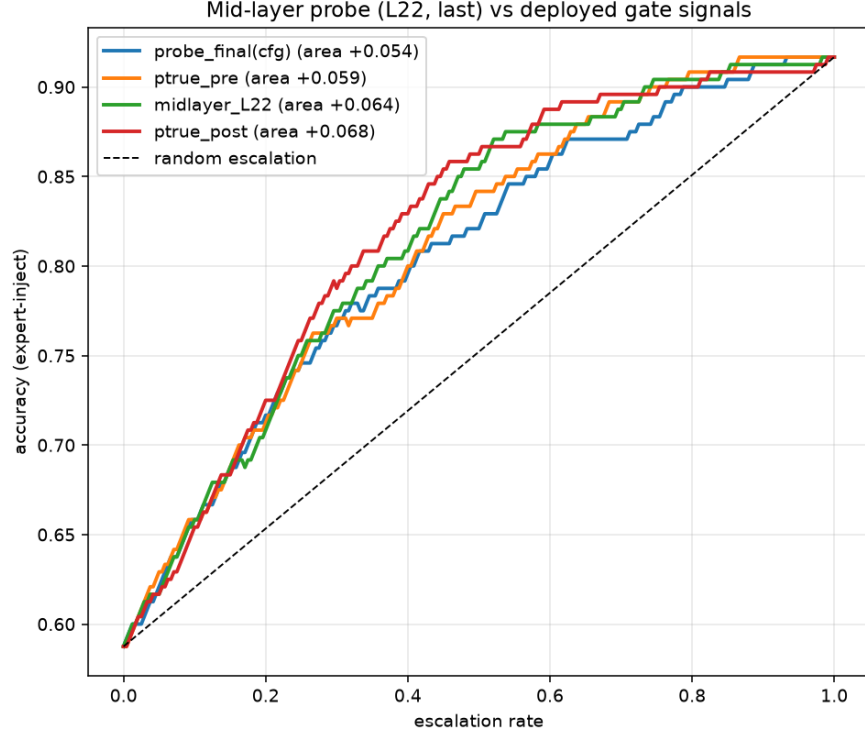


Figure 4: Accuracy vs. escalation rate on the frozen test split for the mid-layer (L22) gate against final-layer probe and $p(\text{True})$ variants.

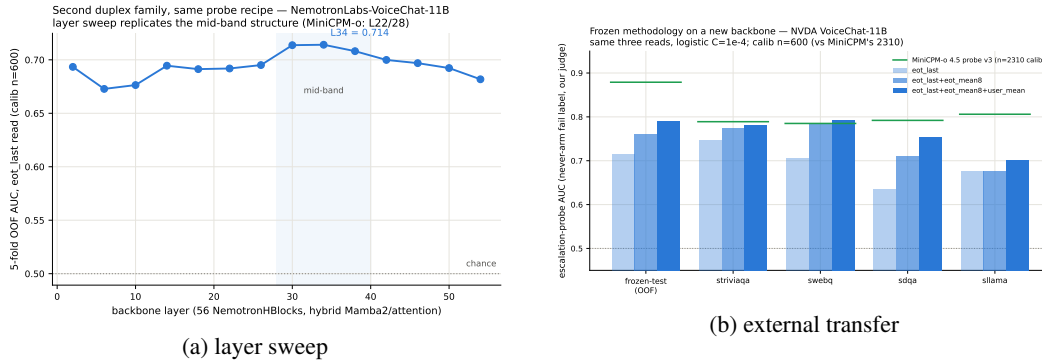


Figure 5: The pre-registered second-family test on NemotronLabs-VoiceChat-11B (hybrid Mamba). (a) The usable read again peaks mid-network (L30–34 of 56). (b) The frozen-recipe probe transfers to the external pools within the MiniCPM v3 AUC band.

input) leaves the cliff unchanged ($L35 = 0.362$) — the operative variable is the modality of the *input* context (text tokens vs. audio embeddings), not the output mode. The duplex fine-tune re-purposed late-layer last-position processing of text-token contexts specifically.

6.2 Cross-modal transfer works — for end-to-end models only

Probes trained on text hidden states and scored on audio hidden states (and vice versa) reveal where the representation becomes modality-shared: early layers are modality-specific (0.54–0.60), and from $\sim 50\%$ depth text→audio transfer plateaus at 0.82–0.87 (0.855 at the deployed L22) on MiniCPM-o 4.5. This replicates on MiniCPM-o 2.6 (onset $\sim 46\%$ depth) and cross-family on

Table 5: Pre-answer introspection under audio input. Trap p_{yes} is the mean stated probability of answering correctly on a pool the model fails 100% (lower is better-calibrated).

model	FT type	trap p_{yes} : text $\rightarrow$ audio	audio $p(\text{True})$ -pre AUC
MiniCPM-o 4.5	duplex	.055 $\rightarrow$.556 (collapse)	.786 (trap dead)
MiniCPM-o 2.6	duplex	.196 $\rightarrow$.345 (compresses to $\sim$.5)	.491 $\approx$ chance
Qwen2.5-Omni	omni-streaming	.279 $\rightarrow$.213 (intact)	.727
Freeze-Omni	frozen + adapter	.165 $\rightarrow$.112 (intact)	(.612; capability caveat)

Qwen2.5-Omni (onset $\sim$ 25%). The decisive control is **Freeze-Omni**: identical frozen Qwen2-7B weights with audio attached via an adapter — and cross-modal transfer is $\approx$ dead (text $\rightarrow$ audio 0.52–0.60, audio $\rightarrow$ text 0.34–0.54). **The modality-shared mid-layer core is created by training the backbone on the modality**; an adapter aligns well enough to converse, but its audio-context hidden states live off the text manifold. Practically: the cheap recipe “calibrate the gate on text data, deploy on speech” is validated for end-to-end models and invalid for adapter-attached ones.

6.3 Verbalized introspection collapses on audio — and why

The mirror-image failure: pre-answer $p(\text{True})$, robust on text (§4.3), collapses under audio input on both MiniCPM duplex generations, while both non-duplex controls stay intact (Table 5).

Three controlled arms pin the mechanism on MiniCPM-o 4.5:

1. *Not perception.* An ASR audit shows the model hears the trap questions fine (WER .074); $p(\text{True})$ computed on the model’s own transcript snaps back to .074 $\approx$ text; the well-heard subset still collapses (.582); $\text{corr}(\text{WER}, p_{\text{yes}}) \approx 0$.
2. *Not a context prior or persona.* Irrelevant filler audio alongside the text question does *not* inflate p_{yes} (.001 — it depresses it everywhere).
3. *Binding.* Duplicating the question as text tokens alongside the audio fully restores introspection (.034). The verbal instance-check runs over **text-token pathways; audio embeddings do not feed it** — while a linear probe reads the same instance information off the same forward pass at 0.93+.

A log-odds decomposition agrees: the audio shift is graded (chat +0.19 $\rightarrow$ trap +4.37), i.e. on audio input verbal self-assessment regresses to type-level priors. The failure also implies its own cheap fix, confirmed twice: re-present the query as text (own transcript or duplication) before asking — “repeat-then-judge.”

6.4 Real-speech validation: not TTS artifacts

Every audio number above uses TTS speech (one voice). We rerun the matched-pair design on 200 real human recordings (VoiceBench SD-QA, USA split; naturally-occurring factoid questions with references), each question presented both as typed text and as the human recording. All three audio-side findings replicate: the modality tax (fail rate .400 $\rightarrow$.450); audio overconfidence (paired p_{yes} shift +.089; on the failure subset .415 $\rightarrow$.581); and the layer structure — early layers dead ($\leq$ L11, .42–.55 across all transfer arms), a sharp rise after $\sim$ 50% depth, audio late layers usable through L35 (.74–.77), and the deployment recipe (text-calibrated probe read on real-speech audio) holding a .76–.80 band at L22–L26 (peak .800 at L25). Magnitudes sit below the matched-TTS .86 as expected: SD-QA content is also out-of-pool, making this transfer across *both* content and modality — a strictly harder test. The text-side math cliff is untestable here (SD-QA contains no math); it never involved synthetic audio and needed no such protection. Residual limitation: scripted read speech in one dialect (ten further dialect splits remain for future robustness work).

Table 6: Mechanism synthesis across training regimes.

	shared mid-layer core	faithful readouts
raw text backbone	(text only)	✓
omni-streaming e2e (Qwen2.5-Omni)	✓built	✓intact
duplex e2e (MiniCPM-o × 2)	✓built	× overwritten (probe: late-layer/text; verbal: audio)
frozen backbone + adapter (Freeze-Omni)	× absent	✓ verbal survives (capability craters instead)

6.5 Synthesis: four quadrants

End-to-end multimodal training *builds* the shared semantic core; duplex-style training additionally *damages* the readouts; omni-streaming gets both right; an adapter alone gets neither (Table 6). There is an elegant symmetry in the two failures: the probe’s readout is text-fragile and $p(\text{True})$ ’s readout is audio-fragile — in both cases the knowledge survives and a *readout* breaks, and in both cases the breakage is duplex-fine-tune-specific. For gate design the consequence is concrete: **the mid-layer last-token probe is the one signal that survives every quadrant that can converse.**

7 The Gate as a System

7.1 Accuracy–cost tradeoff

On the frozen test split ($n = 240$): small-only accuracy **0.588**; big-only (gpt-5.5) **0.917**; big relayed through the small model’s paraphrase 0.879 (relay tax 1–4 points). Every gate beats random escalation at every operating point (Figure 7); the balanced tier at a 33% escalation budget reaches **0.779–0.787**, recovering ~58% of the small→big gap at one third of the big-only cost (\$1.12 per 100 queries big-only).

Figure 7 also draws the baseline the type-shortcut audit (§4.2) demands: a *pool-oracle* router that escalates by each pool’s calibration-split base fail rate (true type labels, no instance-level information, no test leakage). It captures +0.042 of the gate’s +0.054 area over random — the system-level counterpart of the AUC decomposition (0.715 of 0.821), and slightly harsher: in-distribution, the instance-level residual the internal signal adds on top of type routing is +0.012 area (~2 points at the balanced budget) — not significant at $n=240$ (paired bootstrap over test queries, $B=10^4$: 95% CI $[-0.003, +0.027]$, two-sided $p=.13$); in distribution, the gate is statistically on par with the pool oracle. The gate’s case over type routing therefore rests not on an in-distribution margin but on transfer: the type shortcut is exactly what collapses under LOPO (§4.2), while the mid-layer signal survives it.

A *trained* query-feature router closes this argument with data. We fit a RouteLLM-style external router [Ong et al., 2024] under identical data conditions — TF-IDF (word 1–2 gram + character 3–5 gram) → logistic regression on the calibration split’s text labels, seeing only the query surface. In-mix it lands *below* even the pool oracle (OOF AUC .669 vs. .715; tradeoff area +.040 vs. the probe’s +.054, mid-layer +.064, $p(\text{True})$ -post +.068 on the same test set and frozen expert answers); a 24-configuration sweep over feature space and regularization tops out at .689, so the shipped configuration is the query surface’s ceiling, not a tuning artifact. Under LOPO it collapses entirely (held-out-pool AUC .38–.57 vs. the mid-layer probe’s .93 on math) and its mean score on the 100%-fail trap pool is .230 — it *misses* the pool a router most needs to catch. The training receipt makes the point starkly: at the argmax threshold the router’s out-of-fold accuracy on the calibration split, .678, *exactly matches the majority-class rate* (test .613 vs. majority .588) — 360 queries of surface text buy a weak ranking signal (train log-loss .38 vs. out-of-fold .59) and no usable classifier. The internal probes pass the same accounting: at the same threshold, same labels, and same logistic-regression machinery, the text probe’s out-of-fold accuracy is .772 (majority .678) and .779 on test (majority .588), and the live audio mid-layer probe reaches .764/.800 against .592/.512 — real classifiers, not merely rankers; the difference is purely the input representation (Figure 6).

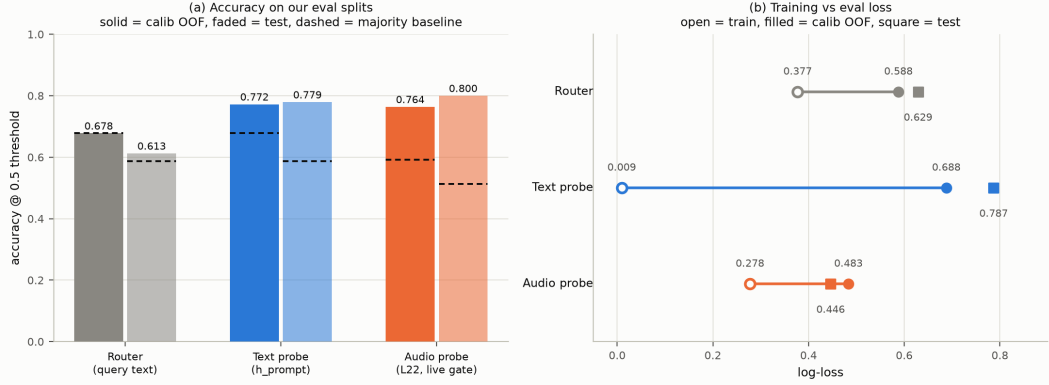


Figure 6: The same training receipt for the external router and the two internal probes (identical labels, identical logistic-regression machinery). (a) Accuracy at the 0.5 threshold on the calibration (out-of-fold) and frozen test splits, against each split’s majority-class baseline (dashed): the router never clears majority; both probes clear it by 9–29 points. (b) Train vs. eval log-loss: the audio probe generalizes cleanly (.278 train / .446 test, the best-calibrated signal); the text probe memorizes in-sample (4096 dims $\gg$ 360 queries) yet its held-out accuracy holds — all reported numbers are out-of-fold or test.

Nor is the baseline data-starved. The identical recipe trained on RouterBench [Hu et al., 2024] — 36,497 public prompts, routing mixtral-8x7b $\rightarrow$ gpt-4 (escalation base rate .432) — reaches out-of-fold AUC .710 in-domain (accuracy .660 vs. majority .568; deferral area over random +.033): a hundred times the data buys .04 of AUC, landing at our pool-oracle’s level and still short of the probe’s .828. Its leave-one-benchmark-out transfer sits at chance on the format-disjoint benchmarks (HellaSwag .502, GSM8K .509, Winogrande .498), reproducing our LOPO collapse on public data, and our calibration-trained router transfers to RouterBench *below* chance (AUC .440). The transfer failure is symmetric: RouteLLM’s *released* preference-trained routers (BERT and matrix-factorization, $\sim$ 100k GPT-4-vs-mixtral preference pairs), scored zero-shot on our pool by their own inference code, reach test AUC only .523/.533 — below even the 360-sample same-data router — and both rank the 100%-fail trap pool *below* the ordinary hard pools. The audio modality narrows but does not close the gap: against the live gate’s audio labels the same recipe reaches OOF AUC .743 / test .814 (gold text; the talker’s own ASR transcript costs it only $-.005$ – $.009$) — stronger than on text labels, because the audio channel makes hard pools fail more uniformly and the type shortcut buys more — yet the audio mid-layer probe still leads every readout (.843/.879 AUC, accuracy .800 vs. .696), and the router still requires the full utterance plus transcription, so it cannot fire mid-stream. Query-feature routing is the type shortcut and nothing else; the standardized LLM-RouterBench protocol [Li et al., 2026] remains future work, but the information-source question is settled under identical data, the ceiling replicates at $100\times$ the data on a public benchmark, and routing knowledge fails to port across talkers in either direction.

7.2 Against the model’s own thinking mode

MiniCPM-o 4.5 ships a built-in `enable_thinking` reasoning mode — self-escalation in place. Table 7 compares deployment policies with measured latencies. **Gated-cloud escalation dominates all-THINK on both axes at every escalation rate.** The reason is mechanistic, not incidental: the gate predominantly flags knowledge/trap failures, which thinking cannot fix — thinking helps math (fail .187 $\rightarrow$.127) but *hurts* chat (.207 $\rightarrow$.233) — so external escalation is necessary, not merely better.

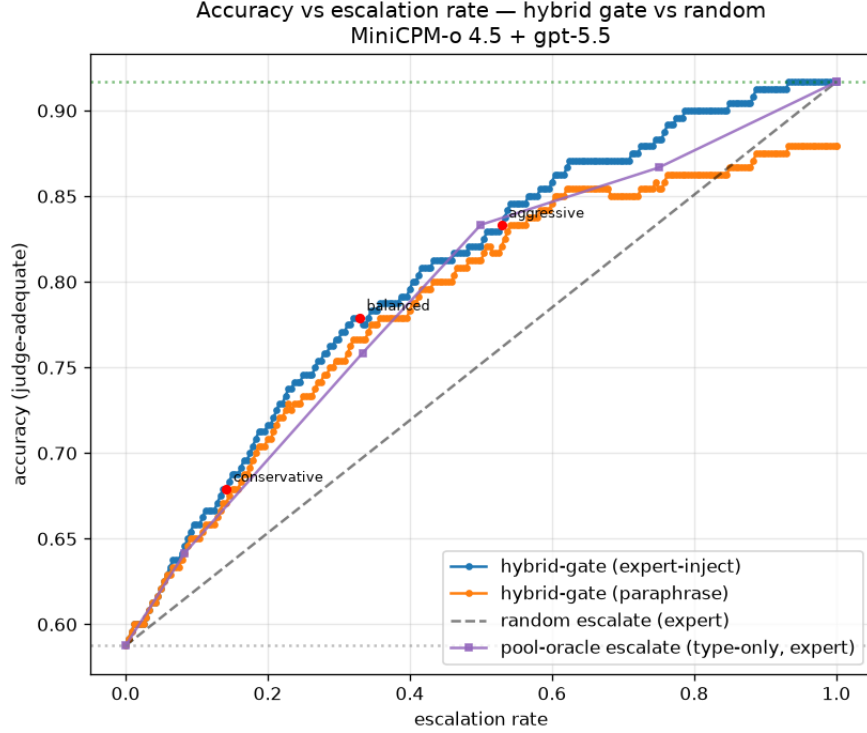


Figure 7: Accuracy vs. escalation rate, frozen test split: gated escalation vs. the random-escalation and pool-oracle (type-only) baselines between the small-only and big-only endpoints. The oracle captures most of the gate’s in-distribution lift (+0.042 of +0.054 area); the gate’s advantage over type routing is transfer, not this margin (§4.2). See also Figure 8 for the $p(\text{True})$ variants.

Table 7: Policy table, frozen test split ($n = 240$). Latency from per-pool measured medians; think and cloud latencies per-query.

policy	acc	lat. mean (s)	P50	P95
fast-only	.588	3.0	3.5	4.2
all-THINK (o4.5’s own)	.637	22.4	17.2	60.1
gated-think @.33	.613	12.1	3.6	47.6
gated-cloud @.15	.688	5.3	3.6	10.1
gated-cloud @.33	.787	6.5	3.6	20.8
gated-cloud @.50	.858	7.0	3.6	24.4

377 7.3 The fork: deciding before the first token

378 The mid-layer probe is one 4096-d dot product on a hidden state the forward pass already computes:
 379 forking a decision branch at layer k costs microseconds and never stalls the main forward. The
 380 design question is at which depth the signal is sufficient. Measured on H100 (CUDA-synchronized,
 381 truncated-forward): the L22 decision lands in **20 ms (text) / 45 ms (audio)**, vs. time-to-first-token of
 382 36 / 68 ms — the decision exists *before decoding starts* and sits far inside the 200–300 ms turn-taking
 383 budget [Stivers et al., 2009]. Per-layer curves (Figure 9) show L22’s hidden state is available at 57%
 384 of prefill wall-time (61% audio) while L22 is simultaneously the in-mix quality peak (OOF .866 >
 385 final layer .835): deciding early costs nothing, and the cloud call can launch while the remaining
 386 ~40% of prefill plus the whole decode still runs. Forking much earlier is not free: pre-50%-depth
 387 layers are both weaker (L10 $\approx$.794) and modality-specific (§6.2); the fork belongs at ~55–60%
 388 depth. For comparison, $p(\text{True})$ -pre costs 39 / 67 ms (an extra short prefill) and $p(\text{True})$ -post costs
 389 a full draft (1.9 / 3.5 s).

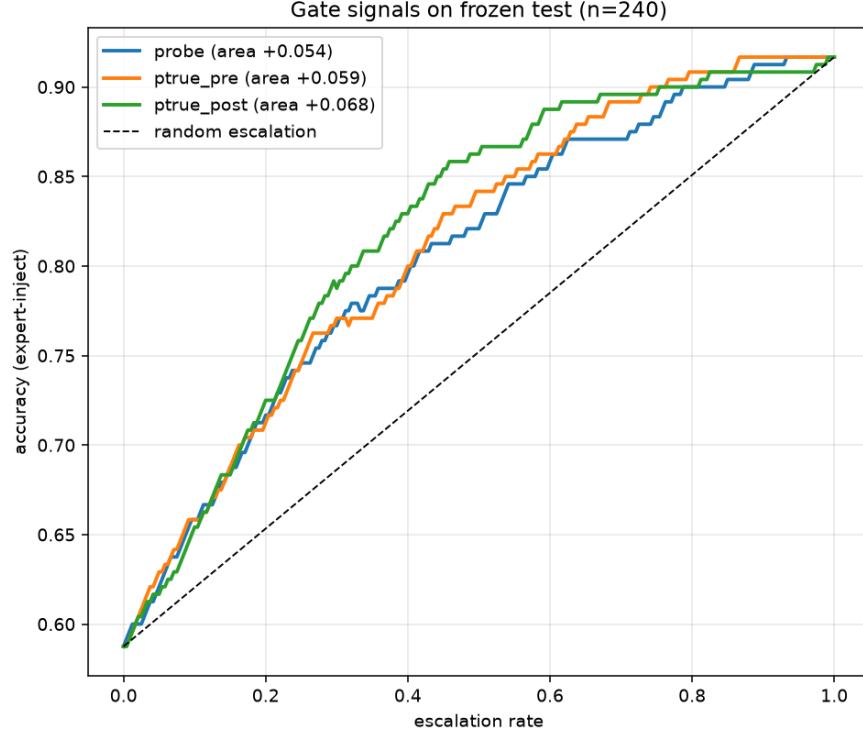


Figure 8: Same tradeoff for the $p(\text{True})$ signals (pre-answer and post-draft).

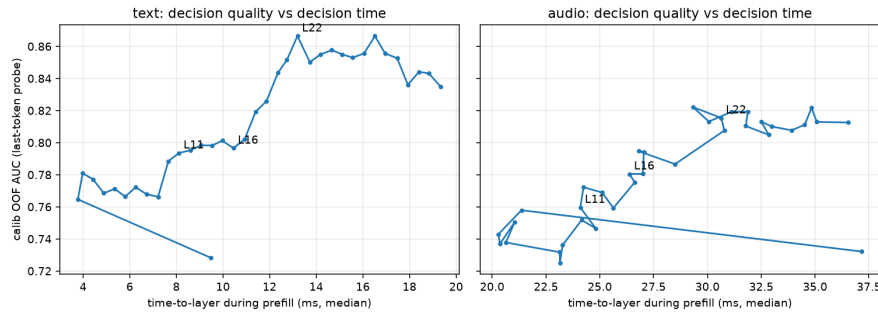


Figure 9: Fork-depth Pareto: per-layer truncated-forward wall-time vs. gate quality. The deployed L22 sits at the quality peak while its hidden state is available at $\sim 57\text{--}61\%$ of prefill wall-time.

7.4 Does the cloud round-trip hide inside the talker's floor time?

If the gate fires at prefill and the cloud call launches immediately, the expert result is ready before the local answer finishes speaking for 40–58% of the whole test mix (audio input buys more time than text). But *gate-escalated* queries skew toward short local answers (trap and knowledge pools; trap overlap ≈ 0), so pure overlap covers only $\sim 20\text{--}31\%$ of escalations at a 33% budget (Figure 10). The residual wait after the local answer ends is P50 2–3 s — one stall sentence — but P90 ~ 27 s, dominated by the expert's own reasoning latency. This argues for a fast-expert tier or streamed partial results in step 2, and it quantifies the stall-phrase budget an injection protocol must cover.

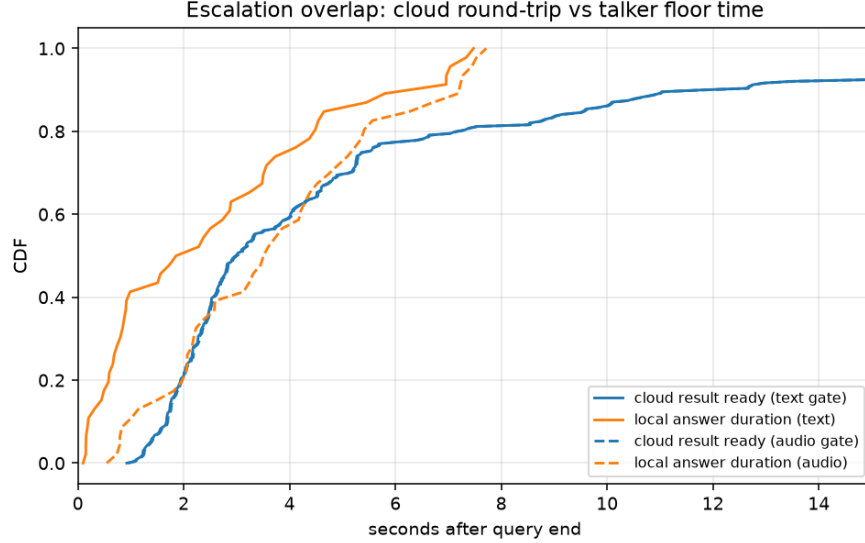


Figure 10: Cloud round-trip vs. local answer duration: probability the expert result arrives before the local answer ends (+slack), whole mix and gate-escalated subset, text and audio.

8 The Gate Goes Live: Streaming Deployment and Result Injection

Everything so far measures signals in replay. This section closes the loop — the original third research question: once the gate has decided the answer belongs to the big model, how does the big model’s result get back into the small model so the *talker* speaks it? We run the full duplex session live: audio streamed in 1 s chunks, the mid-layer gate read during streaming, the expert called mid-conversation, its result injected into the same session. Three questions structure the section: where the streaming gate should read (§8.1), whether the talker will actually say what the expert answered (§8.2), and what the tradeoff curve looks like when every oracle shortcut is removed (§8.3).

8.1 The streaming loop: the end-of-turn read decides, prefetch dies

The offline gate reads h_{prompt} — the last prompt token’s mid-layer state. Streaming has no such token until the turn ends, and per-chunk probe scores are strictly weaker than the offline reference (best chunk statistic AUC .764 vs. offline .843), with a structural failure on trap: short utterances back-load the critical entity into the zero-padded tail chunk, and the mid-stream gate fired on only 1/5 of trap (live) where the offline read caught nearly all. The fix is the **end-of-turn read**: after the last audio chunk, prefill an assistant turn containing a single space and read L22 at the assistant-start position — the streaming analogue of h_{prompt} . This read reaches **AUC .887**, above the offline full-prefill reference (.843), restores trap fire@balanced from .20 to **.90**, costs 22 ms (still before the first output token), and becomes the *only* decision point — escalation-rate thresholds are quantiles of this single score, so the chunk-vs-turn joint-calibration problem disappears.

The remaining role for mid-stream firing would be *speculative* expert prefetch — start the cloud call early, veto at end of turn. We kill this with a measurement before building it. Framing the design as *speculative escalation* — speculative decoding’s draft-verify skeleton [Leviathan et al., 2023] applied to the escalation decision itself, at sentence granularity — its acceptance rate is the probability that an expert answer to the *partial* transcript answers the full question — a wait- k style tradeoff [Ma et al., 2019]. Judged on calibration texts truncated at 25/50/75% of the words, acceptance is **.07/.19/.51**; at the chunk gate’s typical fire point (27–40% of the audio) it is $\sim .2$ – $.4$, i.e. 6–8 of every 10 speculative calls would be wasted. Trap’s acceptance floor (.00/.10/.27) is the same back-loaded-core property that broke the mid-stream probe — one underlying property,

426 measured independently in the hidden states and in the semantics. Prefetch is dropped; the expert
427 starts at end of turn.

428 Two more deployment-honesty decisions close the loop’s design. First, the expert query is the
429 talker’s *own* transcription of the audio (P50 2.2 s, overlapped with the stall phrase), not the dataset’s
430 gold text. Second, expert reasoning effort is **fixed at low**: on the gate-escalated population, effort
431 buys *zero* accuracy (gold-text queries: .85 medium vs. .82 low overall; math 1.00 at both) while
432 doubling latency (P50 5.8→9.1 s, P95 23.9→78.8 s) — escalated failures are knowledge-bound
433 at both scales, mirroring the talker-side thinking result (§7.2). A planned deadline-aware effort
434 policy collapsed, by measurement, to the simplest possible one. The stall phrase itself must be
435 injected as *assistant-role prefill*: the official `teacher_forcing_text` hook continues generating
436 freely mid-stream, so canned filler semantics survive only if the phrase enters as already-spoken
437 assistant speech.

438 8.2 Will the talker say it? The pushback problem, quantified

439 The streaming smoke test surfaced the substantive step-2 obstacle: the talker *pushes back* on injected
440 expert results that conflict with its own reasoning. Before any live sweep, we quantified this channel
441 on 172 live audio sessions: conflicting “expert answers” fabricated by within-pool derangement of
442 reference answers (deterministic, no LLM generation), four framings from neutral (F0) through the
443 deployed authority template (F1), strong-override (F2), and assistant-seeding (F3, teacher-forcing
444 “the answer is: X” into the talker’s own mouth), judged into {comply, pushback, lip-service, other}.
445 Five results:

- 446 1. **The deployed channel is clean.** On the actual live workload — model wrong, correct
447 answer injected — comply = 1.00 ($n=24$). The live curve of §8.3 is not contaminated by
448 relay refusal.
- 449 2. **The same signal that triggers escalation predicts compliance.** Across gate-score quar-
450 tiles, comply rises $.42 \rightarrow .67 \rightarrow .75 \rightarrow .75$; resistance (pushback + lip-service) is .25
451 in the most-confident quartile vs. .08 elsewhere. A model that knows it doesn’t know is
452 willing to listen — the gate and the relay are two uses of one signal. In the high-score half,
453 hard-knowledge (the dominant escalation type) complies 1.00 ($n=13$).
- 454 3. **Authority framing works; forcing words into the talker’s mouth backfires.** Worst case
455 (confident-correct model $\times$ conflicting injection): F0 comply .00, F1 .53, F2 .79 — but F3
456 *seeding* yields comply .25 with lip-service .62: the model speaks the seeded answer, then
457 walks it back in its own continuation. Forced speech is treated as revisable; framing, not
458 ventriloquism, is the mechanism. The deployed loop keeps F1 (F2 is the shelf option: it
459 only helps a rare corner and plausibly raises the swallow rate below).
- 460 4. **Math is silently re-derived.** “Other” concentrates in hard-math: the model neither relays
461 nor disputes — it re-derives and states its own number (comply .57 / silent .26 vs. .66
462 / .20 non-math). A bare numeric answer carries no authority against the model’s own
463 computation chain; harmless when the expert is right, but verbatim-relay claims are off the
464 table for math.
- 465 5. **The swallow rate sets the ceiling.** Model-wrong $\times$ wrong-injection: comply .64, push-
466 back .00. An unconfident model has no internal resistance to a wrong expert — expert
467 quality is the accuracy ceiling of the whole cascade, now quantified.

468 8.3 The live curve, with the deployment price itemized

469 With the channel verified clean, we ran the full live sweep: 240 frozen test queries $\times$ four tiers (never
470 / conservative / balanced / aggressive), each a complete duplex session with the deployed v3 probe
471 (the multi-position L22 refit of §9), scored on *heard accuracy* — the judge sees exactly what a user
472 would have heard (the relay for escalated queries, the local streaming answer otherwise). Labels

Table 8: Live streaming sweep, probe v3, paired bootstrap 95% CIs ($B=10^4$). Top block: speakable subset ($n=218$, headline); bottom: full frozen test pool ($n=240$). Heard = honest end-to-end accuracy; gold-inject = the channel-controlled counterfactual (escalated rows re-scored with the frozen gold-text expert answers); channel cost = their paired difference. * = CI excludes 0; † = CI lower bound at zero.

tier	esc	heard-acc	Δ vs. floor	gold-inject	channel cost
never (floor)	0%	.422 [.36,.49]	—	.422	—
conservative	14%	.528 [.46,.59]	+106* [+05,+17]	.550 [.49,.62]	+023† [.00,+05]
balanced	31%	.592 [.53,.66]	+170* [+11,+24]	.661 [.60,.73]	+069* [+03,+11]
aggressive	57%	.633 [.57,.70]	+211* [+14,+28]	.761 [.70,.82]	+128* [+08,+18]
always (synth)	100%	—	—	.922 [.89,.95]	—
never (full)	0%	.383 [.32,.45]	—	.383	—
conservative	16%	.483 [.42,.55]	+100* [+05,+15]	.525 [.46,.59]	+042* [+02,+07]
balanced	35%	.554 [.49,.62]	+171* [+11,+23]	.654 [.59,.71]	+100* [+06,+14]
aggressive	61%	.596 [.53,.66]	+212* [+15,+28]	.771 [.72,.82]	+175* [+13,+23]
always (synth)	100%	—	—	.917 [.88,.95]	—

are frozen in `frozen_v3_traces.parquet`; all statistics are paired bootstrap over queries on those labels (Table 8, Figure 11).

Headline numbers are reported on the **speakable subset** ($n=218$): a pre-registered input-side fairness filter — decidable from query text and wav bytes alone, never from outcomes — removes 21 formula-LaTeX ids and one digitally-silent wav from the 240. Formula content read aloud by a TTS is destroyed before the system under test ever hears it (no one speaks LaTeX at a voice assistant), and the exclusion is asymmetric in exactly the way the artifact predicts: the gold-inject ceiling barely moves (.917 $\rightarrow$.922) while the aggressive channel gap drops .175 $\rightarrow$.128 — LaTeX rows destroy the speech channel specifically, they are not intrinsically harder for the expert. Rare entities and hard names are deliberately *not* filtered: mishearing them is the phenomenon. Full-pool rows appear in Table 8 for comparison.

The curve rises monotonically and every tier’s gain over the live floor is significant: +.106/+.170/+.211 on the speakable subset, +.21 at 61% escalation on the full pool. Selectivity, however, must be read in the channel-controlled view. In the heard view the higher tiers sit at or below a random-escalation reference drawn to the gold ceiling (aggressive .633 vs. .709): the speech-channel cost (+.128 at that tier) eats the routing margin, so on this mixed pool escalating *at all* helps, but *which* queries the gate picks is masked by the uplink. Remove exactly that channel — the gold-inject view — and the gate clears the same random reference at every arm (+.059/+.083/+.052). The routing signal survives deployment; the binding constraint at high escalation rates is the transcription uplink, not the gate (the external pools of §9, where the channel is milder, show the heard-view advantage directly). One deployment correction fell out of this analysis: re-deriving the tier thresholds as label-free quantiles of the deployment-mix scores (`gate_v3_thresholds_corrected.json`) removes 11.3% of the expert calls at equal accuracy on this pool — the drifted aggressive threshold was a *cost* bug, not an accuracy bug. The interesting content is the *decomposition* of the gap to the offline curve, which the two views separate:

- **Audio + streaming-answer tax (floor .588 $\rightarrow$.383 full pool, .422 speakable).** The live floor stacks the audio modality tax (§6) with a streaming-answer tax: in duplex voice mode the model produces short, speech-styled answers that sometimes plan aloud and stop. This is real deployment behavior — it is exactly why the never-escalate endpoint had to be run live rather than synthesized from chat-mode replays.
- **Speech-channel cost (+.023†/+.069/+.128 on the speakable subset; +.175 at aggressive on the full pool).** On the escalated population, the expert answering the talker’s own transcript scores .585 vs. .871 on gold text ($n=147$ escalated ids). The un-speakable share of this loss is exactly what the fairness filter removes; the deployment-real residue is entity loss on hard-knowledge: prompt-level rescues on the self-transcript are no-ops (a

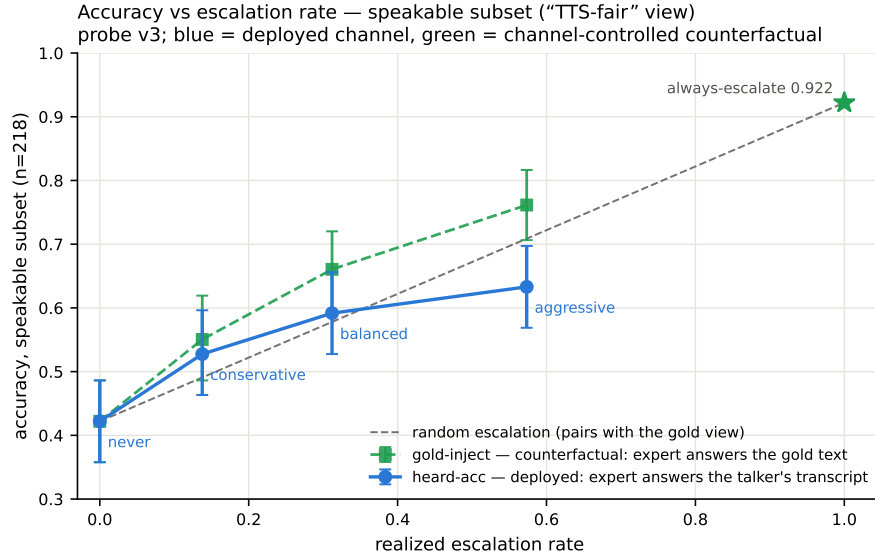


Figure 11: Dual-view live tradeoff on the speakable subset ($n=218$). Blue: honest heard-accuracy of the live duplex system (paired bootstrap CIs). Green: the channel-controlled counterfactual — same live sessions, escalated rows re-scored with frozen gold-text expert answers; its random reference (dashed) uses the same gold outcomes, so it pairs with the green curve only. The two annotated gaps itemize the deployment price: floor-to-floor, the audio + streaming-answer tax; view-to-view, the speech-channel (ASR-distill + relay) cost. Full-pool version: **[TODO: appendix figure (live_dualview)]**.

recognition-errors warning changes nothing; k -best self-transcription trades a small trap gain for a larger knowledge loss) because the damaging errors are internally-coherent entity substitutions with no textual residue to correct from. Replacing the *ears* is different from repairing the transcript: open-weights whisper-large-v3 recovers little (+4 pp), but routing the same ways through a stronger cloud ASR (gpt-transcribe) and handing *that* text to the expert scores .694 vs. .585 (paired McNemar $p=.007$), recovering 38% of the gap to the gold ceiling — a diagnostic result for now: its latency on the critical path and its survival on the external pools are unmeasured **[TODO: add scope note where input-side limits are stated]**. A trace-level decomposition of the escalated heard-failures (first-generation sweep; **[TODO: re-run on v3 traces]**) confirms the loss sits upstream of the relay: 64% are corrupted transcripts (the expert answers the wrong question), 24% are expert misses on clean transcripts (dominated by trap — an expert knowledge limit shared with the gold arm), and only 5 cases show a correct expert answer dropped in relay — consistent with the measured relay compliance of 1.00 on the deployed channel and with expert micro-turn answers being short (median 76 characters), which rules out talker context length as a mechanism. A final controlled arm closes the attribution: sending the *original ways* directly to an audio-capable expert (gpt-audio), with a same-model gold-text control to subtract its capability gap, shows the audio channel itself is lossless for speakable content — trap and easy-fact lose zero crossing it, and the same recordings the talker garbled (“999 – 103” with the minus dropped; “Mustafa Adebayo Balogun”) are heard verbatim by the stronger ear. The synthesis is therefore exonerated along with the relay; the leak is the talker’s own transcription step. We treat ASR robustness as an orthogonal channel property, not a contribution of this system.

531 The replication-noise floor. Live arm accuracies on a 200–250-query pool carry substantial judge-
 532 and-decode replication noise, and we report it rather than narrate inside it: the same query, same
 533 audio, kept *local* in two different arms flips its judge verdict 2.3–18.8% of the time depending on

Table 9: Measured response latency (end of user query $\rightarrow$ answer text complete; seconds), v3 live sweep traces, $n = 240$ per arm (latency is channel-independent, so the full pool is used). Δ columns: loss vs. the never floor at the same statistic. Text-mode pipeline; speech synthesis excluded. P99 on $n = 240$ is indicative only.

tier	mean	P50	P95	P99	Δ mean	Δ P50	Δ P95	Δ P99
never (floor)	4.6	2.0	15.8	17.8	—	—	—	—
conservative	4.6	2.6	12.8	23.0	+0.0	+0.6	−3.0	+5.2
balanced	5.8	3.5	15.9	32.7	+1.2	+1.5	+0.1	+14.9
aggressive	6.6	4.4	18.5	33.0	+2.0	+2.4	+2.7	+15.2

pool (this pool: .155); repeated *escalation* flips 0.7–10.6%. The implied paired SE on an arm-vs-arm accuracy delta is .009–.028, so any live delta under ~ 3 points is undetectable in a single sweep — accordingly, no claim in this paper rests on a sub-3-point live difference; probe comparisons rest on offline AUC and matched-rate precision, which are far tighter statistics on the full score distribution. **[TODO: noise_audit methods figure — appendix]**

What escalation costs in wall-clock time. The same frozen traces carry per-session timers, so the latency price of each operating point is measured, not modeled (Table 9). The gate itself is free at this scale: the end-of-turn read and the stall prefill cost ~ 30 ms each at P50; the expert round-trip (GPT-5.5, low effort; P50 4.8 s, P95 20.8 s) dominates escalated-session wall time (escalated-session P50 5.1–7.5 s vs. local 1.9–3.0 s). The consequence is a specific distributional signature: at the balanced tier the *mean* response time rises only +1.2 s and the P50 +1.5 s over the never floor, but the P99 nearly doubles (17.8 $\rightarrow$ 32.7 s) — escalation is cheap on average and expensive exactly in the expert-latency tail ($n = 240$ per arm, so P99 rests on the two worst sessions and is indicative only). The conservative tier even *shortens* the tail (P95 15.8 $\rightarrow$ 12.8 s): its few escalations displace exactly the longest local decodes, the same median-of-a-mixture mechanism analyzed on the external pools (§9). Figure 12 replays three real balanced-arm sessions from their recorded timers, showing the three shapes a session takes: the expert beating the stall phrase (seamless handoff), the expert outliving it (dead air — the tail case above), and the gate staying closed (first token in 0.4 s). Figure 13 joins the accuracy and latency readouts into the exchange rate actually paid at the median (speakable subset): +10.6 points of heard accuracy for +0.3 s (never $\rightarrow$ conservative), +6.4 for +1.0 s (conservative $\rightarrow$ balanced), and +4.1 for +1.1 s (balanced $\rightarrow$ aggressive). The first transition is both the steepest and the cheapest per second, for two measured reasons: escalation displaces local decodes that average 4.6 s (and the conservative tier’s escalations are precisely the slowest of them, hence its P95 shortening above), while the marginal escalations added at higher tiers reach easier queries whose experts return faster (escalated-session P50 7.5/6.9/5.1 s across the three tiers).

8.4 Signal boundary: a third blind spot

A fairness challenge to the trap pool — SimpleQA entities are hard to *hear*, conflating knowledge difficulty with transcription difficulty — motivated auditioning a transcription-fair failure pool: false-premise questions [Hu et al., 2023], every word common by design ($n = 60$, criterion fixed before data selection). The audition fails *as a pool* and succeeds as a finding: the small model fails substantially (.63 text / .47 audio) and the expert would help (.80 adequacy), but the end-of-turn gate is **blind** to these failures — fire@balanced .18 (vs. trap’s .90), with essentially no score separation between failed and passed answers (.568 vs. .507). A model answering along a false premise does not “feel” unable: the mid-layer signal reads *missing knowledge*, not *broken question*. This is the third documented blind spot, after easy-chat failures and decode-time math: the pre-answer signal is specifically a *retrieval-failure* detector. FalseQA therefore enters the signal-boundary analysis, not the tradeoff pools.

A boundary case that data can move: real-time queries. A fourth query class surfaced in live demo use: questions whose answer is unknowable at any training cutoff (“what is NVDA trading

Three live sessions, timestamped (balanced arm; every event time is a recorded live-loop timer)

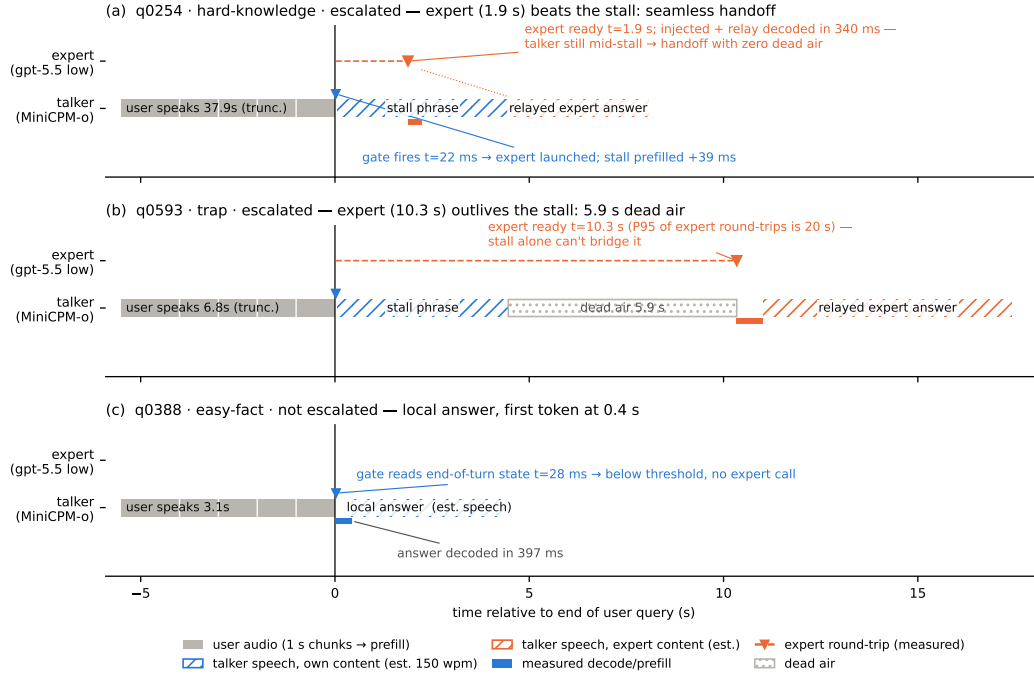


Figure 12: Three real live sessions (balanced arm), replotted from their recorded timers. (a) The expert returns at 1.9 s, while the talker is still voicing the stall phrase: the answer is injected mid-stall and the handoff has zero dead air. (b) The trap-pool case: the expert takes 10.3 s, the stall ends first, and 5.9 s of dead air follow — the P99 signature of Table 9 as a single session. (c) The gate stays closed and the local answer’s first token arrives at 0.4 s. Hatched speech bars are the one estimated quantity (150 wpm; the live loop emits text); every event timestamp is measured.

at right now”). Unlike false premises, the gate is not blind here — v3 scores sit *on* the balanced threshold, flipping with the TTS render (.624 vs. .693 for the same question) — the class was simply absent from every (static-knowledge) training family. Extending the calibration pool with FreshQA [Vu et al., 2023] closes it: fast-changing questions (153, labeled escalate a priori — the current value is unknowable by construction, so a judge against a stale gold would only add noise) plus never-changing in-family controls (150, standard judge labels, fail rate .47). Refit with the frozen v3 feature config, the v4 probe fires on held-out fast-changing questions at .80/.93/1.0 across tiers (v3: .57/.80/1.0) while never-changing held-out rates and frozen-test AUC are unchanged (.877 vs. .879). One calibration subtlety: tier budgets must remain quantiles of the *deployment-mix* scores; quantiling over the augmented train mix (123 extra all-positive rows) silently raised the balanced threshold from .680 to .737, spending the new capability on its own threshold shift. Escalation only pays for this class if the expert can reach live data — the demo’s expert therefore gains a web-search tool, while all measured arms keep the tool-free expert.

9 Transfer Without Training: The Frozen Gate on Six Public Speech Benchmarks

Everything so far is measured on our internal pool, and §4.2 showed exactly why that is not enough: in a mixed pool, most of a router’s apparent margin can be the pool/type shortcut. This section is the out-of-distribution test the audit demands. We take the gate *frozen* — probe weights, feature configuration, and layer choice fit once on the internal calibration mix — and run the complete live streaming loop of §8 on six public speech-QA benchmarks: Speech TriviaQA, Speech Web

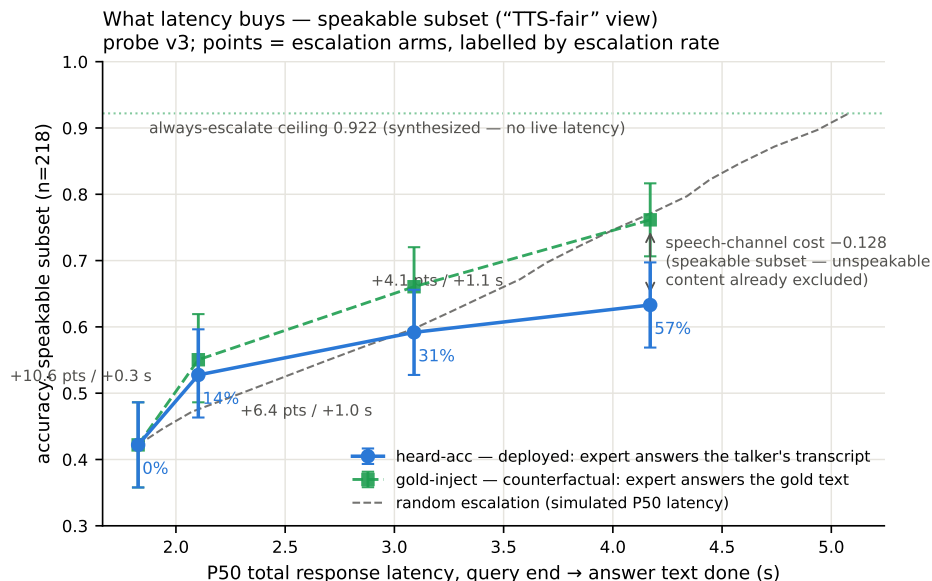


Figure 13: The live accuracy–latency tradeoff on the speakable subset: per-arm P50 response latency against accuracy with 95% paired-bootstrap CIs (Table 8), in the honest heard view (blue) and the channel-controlled gold-inject view (green) of the same sessions — rescoring changes outcomes, not timing, so the two views share their x positions. Segment labels give the median exchange rate on the heard curve; the dotted curve is the simulated-P50 random-escalation reference. The always-escalate ceiling is synthesized from frozen gold answers and has no live latency. Median view — the expert P99 tail is priced in Table 9. Full-pool version: **[TODO: appendix (pareto_latency)]**.

593 Questions, Llama Questions, and Reasoning QA from OpenAudioBench **[TODO: cite: OpenAu-**
594 **dioBench; Web Questions (Berant et al. 2013); Llama Questions (Nachmani et al.)]** [Joshi et al.,
595 2017], SD-QA (real dialectal human speech, not TTS) [Faisal et al., 2021], and VoiceBench Al-
596 pacEval [Chen et al., 2024, Taori et al., 2023]. All pools use the benchmarks’ official pre-rendered
597 audio — none of it passes through our TTS — and none of their queries ever touches probe training.
598 The deployed probe (v3) is the same L22 end-of-turn read plus two additional zero-latency posi-
599 tions (a rolling 8-token tail and the running user-audio mean), refit on a calibration mix widened to
600 2,310 queries drawn from public families (SVAMP, HotpotQA, PopQA, MMLU, TruthfulQA-style
601 traps) that are disjoint from — and deduplicated against — the six evaluation benchmarks; the pre-
602 registered guard held with headroom (frozen internal test AUC .860→.879, and feature selection
603 never saw the externals). The only per-pool adaptation is *label-free*: escalation-tier thresholds are
604 quantiles of the probe’s own score distribution on each pool, a normalization the score scale forces
605 (the aggressive threshold is .111 on Llama Questions vs. .603 on SD-QA — a 5× shift for the same
606 probe) and one that consumes no labels, no answers, and no judge calls.

607 Scoring is each benchmark’s own where a protocol exists: the three OpenAudioBench pools use
608 the official judge verbatim (alias-match grading), AlpacaEval uses VoiceBench’s official 1–5 judge,
609 and SD-QA / Reasoning QA (which ship no replicable protocol) use our reference-anchored judge.
610 Under the official judge our offline chat-mode control *reproduces* the anchors — .712 vs. the official
611 .755 on Speech TriviaQA (250-item subsample + protocol) and .716 vs. .702 on Web Questions —
612 so the official lines in our figures are on-scale, and the live floor’s remaining gap to them is fully
613 accounted (striviaqa: −.048 streaming-loop tax, −.043 subsample/protocol; nothing unexplained).

Table 10: The frozen gate on six public speech benchmarks, live streaming loop, v3 probe, realized escalation rates 15/30/50%. Judge: OAB = the benchmark’s official OpenAudioBench judge; VB = VoiceBench’s official 1–5 judge; ours = reference-anchored (no official protocol exists). Ceiling = GPT-5.5 on gold question text under the same judge; AUC = v3 probe score vs. never-arm local-fail label; official = MiniCPM-o 4.5’s reported number (offline chat mode; our own chat-mode control reproduces it, see text). Per-arm live accuracies carry a replication-noise floor of $\pm .02-.03$ (§9.3).

pool	judge	n	escalation tier				ceiling	AUC	official
			0%	15%	30%	50%			
Speech TriviaQA	OAB	250	.664	.732	.800	.860	.972	.789	.755
Speech Web Questions	OAB	250	.57	.628	.680	.732	.844	.785	.702
Llama Questions	OAB	250	.84	.904	.904	.948	.928	.806	—
SD-QA	ours	200	.51	.600	.720	.795	.930	.792	—
Reasoning QA (zh)	ours	202	.59	.653	.713	.772	.871	.683	—
AlpacaEval (1–5)	VB	199	3.99	3.96	4.23	4.35	4.96	—	4.8

9.1 Accuracy: the mechanism ports, and the anchors fall

Table 10 is the headline: the frozen gate transfers mechanically everywhere — thresholds fire at sane rates, every curve rises monotonically, and the 50%-escalation arm lifts the live floor by +.11 to +.29 across the five QA pools. Three anchor results:

Routing beats scaling the speech backbone. On Speech TriviaQA the 9B duplex model plus the frozen gate at 50% escalation scores **.860 live**, against .755 for MiniCPM-o 4.5’s own official offline number and **.629** for Qwen3-Omni-30B-A3B ($3.3\times$ the parameters, offline chat mode) [TODO: cite: Qwen3-Omni / Kimi-Audio comparison rows (official OpenAudioBench table; Kimi-Audio scores .419 here)]. On Web Questions the gated system clears the official number (.732 vs. .702) and lands just under Qwen3-Omni (.749).

On Llama Questions, selective escalation matches the expert at half the cost. The pool is single-type (short English factoids), so no type shortcut exists to exploit, and the local model is strong (floor .84). The probe still splits it cleanly: at 50% the kept-local half scores **.976** vs. **.696** on the escalated half ($z=6.46$) — genuine per-query discrimination — and the expert’s entire advantage concentrates where the probe sends it (escalated half .696 $\rightarrow$.888, McNemar $p<.0001$) while buying nothing measurable on the kept half (.976 vs. .968, $p=1.00$). Selective at 50% reaches .948 vs. .928 for always-escalate (directionally above, $p=.125$ — both near ceiling): escalating everything spends twice the expert calls to buy nothing measurable over the gate.

Router quality must be read as AUC plus matched-rate precision, not raw precision. Precision at a fixed escalation rate is capped at base-fail/rate: on Llama Questions (base fail .16) the cap at 50% is .32, and the v3 gate’s matched-rate precision of .304 is **95% of the theoretical maximum** (recall .93); quoting “.30 precision” without the cap would misread the strongest pool as the weakest. The scale-free readout is transfer AUC against the never-arm fail label: .68–.81 on every external pool (Table 10) vs. .879 on the internal held-out test — the mid-layer signal ports as a generic difficulty signal, with a near-uniform matched-rate precision lift of $\times 1.33-1.35$ across pools rather than pool-specific structure. The cleanest evidence that the signal is difficulty, not surface, is cross-lingual: the calibration expansion is *English-only*, yet it moves Reasoning QA — a *Chinese* execution-reasoning pool — from AUC .621 to **.683** (+.062), the largest per-pool gain of the refit. Per the replication-noise analysis below, the v3 evidence is this offline AUC shift; the corresponding live-arm deltas (+.010–.030) are individually within noise and we do not narrate them.

9.2 Latency: the probe escalates the slowest decodes first

The latency curves of Figure 14 have a shape worth explaining honestly, because at first sight it looks like a measurement error: on the easy pools the per-arm P50 is *non-monotonic* — on Llama Questions it *falls* from 1.52 s (never) to 1.17 s at 15% escalation before rising again (1.60/1.42 s at

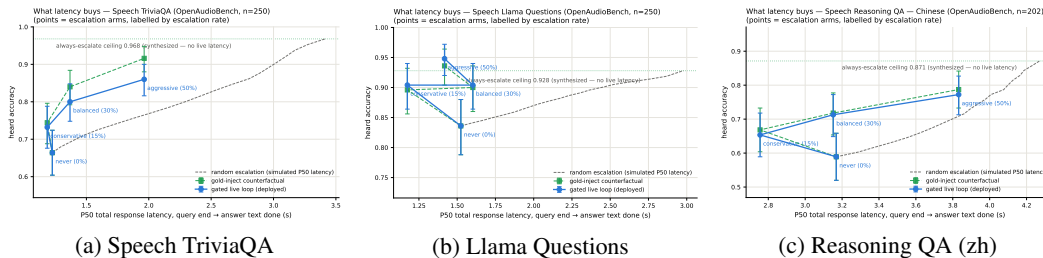


Figure 14: Accuracy vs. per-arm P50 response latency on three external pools (frozen gate, live arms; official judge on a–b, ours on c). The random reference simulates a random mixture at each rate from per-id *measured* latencies (400 sims $\times$ 21 rates). On (a) and (b) the gated curve dominates random in *both* axes: on Speech TriviaQA random needs ~ 3.4 s at the median to reach the accuracy region the gate reaches at 2.0 s, and on Llama Questions random at 50% sits at ~ 2.2 s/.88 vs. the gate’s 1.42 s/.948. On (c) escalation truncates the latency *tail* (P90 13.25 $\rightarrow$ 10.94 s). Remaining family figures: **[TODO: appendix: dualview + pareto for all six families]**.

30/50%). This left-fold is a verified median-of-a-mixture mechanism, not noise: the probe preferentially escalates exactly the queries whose *local* decode is longest. On Speech TriviaQA the 38 queries escalated at 15% had a never-arm local P50 of 1.711 s vs. 1.198 s for the 212 kept — and, measured in the never arm, local accuracy .32 vs. .73: the small model’s signature failure mode on retrieval pools is hedging at length, so wrong answers are slow answers (decode time and answer length are the same variable, $r=-.97$), and a wrongness-selecting probe strips the slow tail of the local mixture as a side effect. Controls: the probe does not key on length directly ($\text{corr}(\text{score}, \text{length}) = +.05$; conditioned on wrongness the score-length coefficient is ≈ 0), and on SD-QA — where wrong answers happen to be short and confident — the same probe selects wrongness identically and produces *no* fold. Figure 15 decomposes the effect: the fold lives only in the median — the *mean* is monotonic (1.65/1.81/1.96/2.06 s vs. P50 1.52/1.17/1.60/1.42 s) — and the kept-local P50 falls 1.52 $\rightarrow$ 0.96 $\rightarrow$ 0.86 $\rightarrow$ 0.43 s as the probe strips slow decodes while escalated rows pay a flat ~ 3 s expert round-trip. We report the mean alongside the median for exactly this reason; the honest statement is that on these pools escalation buys accuracy *and*, at the median, speed. The individual dip depth is within noise (striviaqa -41 ms, bootstrap CI $[-285, +254]$ ms) but its sign is deterministic — identical probe scores select the identical escalated set on every rerun.

On Reasoning QA the shape inverts usefully: P50 rises gently but the *tail improves* — P90 drops from 13.25 s to 10.94 s under escalation. The mechanism is that the local model’s chain-of-thought is *spoken*, so every reasoning token enters the response clock (escalated-set local decodes P50 4.38 s, tail 13 s+), while the expert’s chain is hidden: its median completion is 144 tokens of which only 57 characters return, behind a flat ~ 3.7 s round-trip. On reasoning pools the gate replaces the longest spoken chains with short round-trips — escalation truncates the latency tail rather than fattening it.

9.3 Where transfer fails, and what the live curves can and cannot show

AlpacaEval is a negative result, mechanism measured. On VoiceBench AlpacaEval the gate does *not* beat random: aggressive 4.35 vs. random-at-50% ≈ 4.45 . The failure is selection, not escalation — queries the gate selected score 3.90 on the never arm vs. 3.98 for those it skipped (no discrimination), while escalated rows still gain (3.90 $\rightarrow$ 4.71) purely because GPT-5.5 writes better long-form answers; random picks would harvest the same gain. Open-ended instruction following has no “the model lacks this fact” event for a retrieval-failure detector to read, and the score range is compressed (1.0 point vs. 40 accuracy points on our pool). Two scale caveats make the absolute level honest: the official 4.8 is an *offline chat-mode* number — our chat-mode control reproduces it at 4.86 — and the entire chat-vs-live gap (4.86 vs. 3.99) is the duplex loop’s spoken-reply style plus a 512-token cap, so only the live arms are comparable to each other.

Web Questions and SD-QA are flat under the v3 refit, offline (AUC $+0.006/+0.017$) and live alike. On Web Questions a specific share of the blame is label fidelity rather than the model: its strict

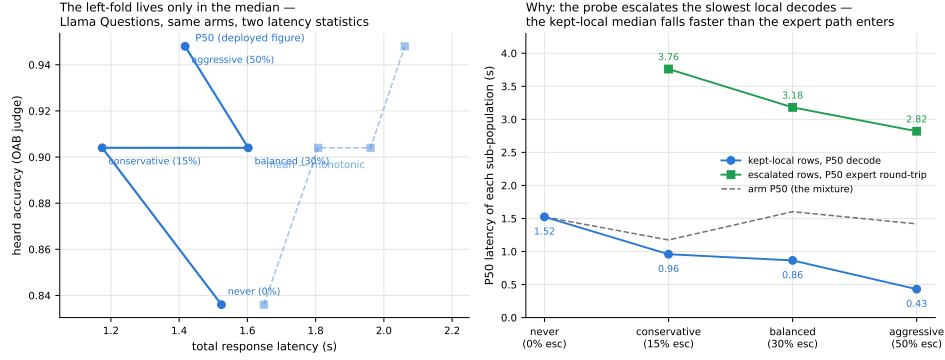


Figure 15: Decomposition of the Llama Questions latency fold. Left: the fold exists only in the median — the mean is monotonic. Right: kept-local P50 falls 1.52→0.43 s across tiers as the probe strips the slowest local decodes, while escalated rows pay a flat ~3 s expert round-trip; the arm median folds left and then rises as the expert mass reaches it.

Freebase-alias grading decouples “wrong” from “uncertain” — the wrong-answers-are-longer signature is null there for every model we measured, including GPT-5.5 on gold text — so a portion of its labels is protocol-wrong that no state-reading probe can predict, and even the gold-text expert ceiling is .844.

The deployed view understates selection on easy pools. The external channel cost is small — the aggressive-arm heard-vs-gold-inject gap is .040/.024/.025 on striviaqa/swebq/sdqa [TODO: verify: gold-inject gaps on the v3 traces; quoted values are from the v2-probe sweep] vs. .175 on the internal pool (.128 on its speakable subset, §8.3), exactly as predicted for short open questions with no option walls — but on easy homogeneous pools even a small tax eats a thin per-query margin, and part of the deployed-view “gate ≈ random” on Speech TriviaQA is channel cost: in the channel-controlled gold view the gate clears its paired random line (+.044 at the aggressive arm). SD-QA meanwhile carries the one property no TTS pool has — real dialectal human speech — and the escalated subset still converts at .82–.84 with expert P50 3.1–3.4 s: the payload survives real speech.

Replication-noise discipline. Re-running the same query with the same audio through the live loop flips the judge verdict 2.3–18.8% of the time depending on pool (Llama Questions .023, SD-QA .188), implying a paired SE of .009–.028 on any arm-vs-arm accuracy delta. Against that floor, all 18 per-arm live v2→v3 deltas are individually non-significant (the largest, striviaqa balanced +.036, gives McNemar $p=.16$). We therefore treat sub-3-point live deltas as unresolvable in a single sweep and rest every probe-improvement claim in this section on the variance-reduced instruments — offline AUC over the full score distribution and matched-rate precision — with the live curves serving as the end-to-end existence proof, not the measurement.

10 Discussion

10.1 Design consequences: the two-stage gate

The evidence converges on a two-stage design:

- **Stage 1 (prefill, effectively free):** a mid-layer (~60% depth) last-token probe — 20/45 ms, before the first token, and the only signal that survives duplex fine-tuning *and* both input modalities (§6.5). Open engineering item: score-scale calibration for threshold (rate) transfer.
- **Stage 2 (optional, post-draft):** $p(\text{True})$ -post on the drafted answer — the strongest single signal (0.899); the draft doubles as the distilled escalation query and as the fallback an-

713 swer. On audio input, never ask over raw audio context: run $p(\text{True})$ on the model’s own
714 transcript (“repeat-then-judge,” §6.3).

715 10.2 Positioning vs. query-side routing

716 Query-side routers [Ong et al., 2024, Ding et al., 2024, Chen et al., 2023] decide before generation
717 from query features, and the big model then answers the user directly. Neither transfers to full-
718 duplex voice: the user talks to one voice whose session state lives in the small model, and query-side
719 features are precisely the “type shortcut” our audit quantifies (pool oracle 0.715 vs. probe 0.822 in-
720 distribution) and LOPO breaks. The signal must come from the answering model itself — consistent
721 with Li et al. [2026]’s attribution of router failures to model-recall gaps. A head-to-head compari-
722 son of query-feature routers vs. internal-signal gates under distribution shift, on their standardized
723 protocol, is natural future work.

724 10.3 Result injection: resolved by the same signal

725 Step 2 — returning the thinker’s result into the live duplex session — initially looked like a research
726 problem of its own: the streaming smoke test showed the talker pushing back on conflicting injected
727 results. The quantified answer (§8.2) is that for *gated* escalations the problem largely dissolves, and
728 for a principled reason: the same low-confidence state that fires the gate predicts relay compliance
729 (comply .42 → .75 across gate-score quartiles; 1.00 on the deployed wrong-model × correct-expert
730 workload). The gate and the relay are two uses of one signal. Where the state is confident, framing
731 — not ventriloquism — is the mechanism (authority framing .53–.79 vs. seeding’s backfire), and
732 the residual ceiling is the swallow rate: an unconfident talker relays a *wrong* expert at .64 with zero
733 pushback, so cascade accuracy is bounded by expert quality.

734 10.4 Three failure species, one perceivability axis

735 The accumulated blind-spot results organize into a taxonomy of *why* a model fails, ordered by
736 whether the failure is an observable event in the model’s own processing. **Retrieval-type** failures (“I
737 don’t know this entity”) produce an observable empty lookup at prefill — pre-answer signals work
738 (trap fire .90). **Execution-type** failures (“stuck mid-solution”) only come into existence during de-
739 coding — decode-time signals see them (math entropy@16 .860) and pre-answer signals cannot.
740 **Metacognitive-type** failures (“doesn’t know the question itself is broken”) never occur as an event
741 anywhere in the model’s experience — like a schoolkid confidently “verifying” Goldbach’s con-
742 jecture, the model answering a false-premise question executes smoothly along the wrong frame, and
743 every internal signal is blind (FalseQA fire .18, no score separation; §8.4). This mirrors the human
744 feeling-of-knowing literature, which finds metacognition accurate for memory retrieval and unreli-
745 able for insight problems. The taxonomy explains the two-stage gate architecture (§10.1 rationale:
746 prefill probe for species 1, decode monitor for species 2), scopes the claim — the pre-answer sig-
747 nal is a retrieval-failure detector, not a general oracle — and notes that query-surface routers miss
748 species 3 as well.

749 10.5 Demonstration system

750 The measured pipeline runs as an interactive web demo: live microphone audio streams to the duplex
751 model on a single H100, the end-of-turn probe read costs ~54 ms against the balanced threshold, and
752 escalated turns stall, consult the expert, and *speak* the relayed answer through the model’s own TTS
753 head. Two demo-only extensions go beyond the measured arms: the expert gains a web-search
754 tool — real-time queries such as stock prices, which the v4 FreshQA refit (§8.4) fires on at .80–1.0
755 across tiers, then return live values no cutoff-bound model can produce — and the loop accepts
756 *backchannel-aware barge-in*: sustained speech over the talker ducks playback, a ~1 s ASR pass
757 classifies the burst, and backchannels (“ok”, “mm-hm”; an English+Chinese lexicon) return the floor
758 with zero loss while anything else aborts generation and seeds the next turn. All measured arms in

759 this paper keep the tool-free expert and the turn-based loop; the demo will be made available upon
760 publication.

761 10.6 Limitations

762 (i) $p(\text{True})$ prompt sensitivity: one phrasing each (pre/post). (ii) The audio arm is validated on
763 real dialectal speech (SD-QA), but scripted read speech and one dialect split; spontaneous conver-
764 sational speech (disfluencies, self-repair) remains uncovered. (iii) Mid-layer probe threshold trans-
765 fer needs deployment-grade calibration; AUC/area claims are unaffected. (iv) The audio side of
766 the introspection-collapse finding is MiniCPM-scoped (two generations of one family); a new open-
767 weight full-duplex family is a pre-registered prediction test — it should show the late-layer text-input
768 cliff. (v) The frozen test split is reused across signal comparisons (signals were pre-specified; a fresh
769 test set would be cleaner). (vi) Labels come from a single judge family, bounded by a stronger re-
770 judge ($\text{deltas} \leq 0.010$) but without human labels. (vii) The live sweep (§8) is one session per query
771 per tier; the conservative tier’s gain over the floor is not significant at $n = 240$. The measured live
772 sweep ran with text-mode talker output (`generate_audio=False`): the relay is validated at the
773 utterance level, and its latencies are text-complete — a conservative accounting, since in the spo-
774 ken system the stall filler plays the moment the gate fires. The released demo now speaks with the
775 model’s native TTS head (talker + Token2wav vocoder, 24 kHz; first answer audio $\sim 0.9\text{--}1.4$ s after
776 generation start; the TTS template token enters after the end-of-turn read, so the probe context and
777 all frozen thresholds are unchanged). The demo also supports barge-in with backchannel awareness
778 — speech over the talker first ducks playback, a fast ASR pass then either returns the floor (“ok”,
779 “em”) or commits the interrupt (“stop”, new content), which aborts generation at chunk granularity
780 and seeds the next turn. This is a lexicon-level approximation of what a native duplex listen/speak
781 head decides from hidden state; that head remains unused, since the probe is calibrated on simplex
782 streaming states, and its validity under the true duplex loop is an open experiment.

783 11 Conclusion

784 Zero-training self-knowledge in a small full-duplex model exists, is strong, and is modality- and
785 layer-fragile in systematic, duplex-fine-tune-induced ways. The standard readouts — final-layer
786 probes and verbalized self-evaluation — each have one blind modality, and both blind spots are
787 created by the duplex fine-tune; the mid-layer representation underneath survives everything we
788 tested. Read there, the resulting gate decides before the first output token, transfers from cheap text
789 calibration to speech on end-to-end models, and as a system dominates the model’s own thinking
790 mode on both accuracy and latency. And when the thinker’s answer comes back, the same signal
791 that fired the escalation predicts that the talker will actually say it: run live end-to-end, the gated
792 duplex system climbs monotonically from its honest streaming floor ($.422 \rightarrow .633$ at 57% escalation
793 on the speakable subset), with the remaining gap to the offline curve itemized into named, measured
794 taxes rather than left as mystery — and the frozen gate transfers unchanged to six public speech
795 benchmarks, where at 50% escalation it lifts the 9B system past models $3\times$ its size. Gate and relay
796 are one mechanism; what remains is engineering the channel around it.

797 References

- 798 Pranjali Aggarwal, Aman Madaan, Ankit Anand, Srividya Pranavi Potharaju, Swaroop Mishra, Pei
799 Zhou, Aditya Gupta, Dheeraj Rajagopal, Karthik Kappaganthu, Yiming Yang, Shyam Upadhyay,
800 Manaal Faruqui, and Mausam. AutoMix: Automatically mixing language models. In *Advances*
801 *in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv:2310.12963.
- 802 Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier
803 probes. *arXiv preprint arXiv:1610.01644*, 2016.
- 804 Amos Azaria and Tom Mitchell. The internal state of an LLM knows when it’s lying. In *Findings*
805 *of the Association for Computational Linguistics: EMNLP 2023*, 2023. arXiv:2304.13734.

806 Collin Burns, Haotian Ye, Dan Klein, and Jacob Steinhardt. Discovering latent knowledge in lan-
807 guage models without supervision. In *International Conference on Learning Representations*
808 (*ICLR*), 2023. arXiv:2212.03827.

809 Lingjiao Chen, Matei Zaharia, and James Zou. FrugalGPT: How to use large language models while
810 reducing cost and improving performance. *arXiv preprint arXiv:2305.05176*, 2023.

811 Qian Chen, Yafeng Chen, Yanni Chen, et al. MinMo: A multimodal large language model for
812 seamless voice interaction. *arXiv preprint arXiv:2501.06282*, 2025.

813 Yiming Chen, Xianghu Yue, Chen Zhang, Xiaoxue Gao, Robby T. Tan, and Haizhou Li. VoiceBench:
814 Benchmarking LLM-based voice assistants. *arXiv preprint arXiv:2410.17196*, 2024.

815 Yunfei Chu, Jin Xu, Qian Yang, Haojie Wei, Xipin Wei, Zhifang Guo, et al. Qwen2-Audio technical
816 report. *arXiv preprint arXiv:2407.10759*, 2024.

817 Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, et al.
818 Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021. GSM8K.

819 Mike Conover, Matt Hayes, Ankit Mathur, Jianwei Xie, Jun Wan, Sam Shah, Ali Ghodsi, Patrick
820 Wendell, Matei Zaharia, and Reynold Xin. Free dolly: Introducing the world’s first truly open
821 instruction-tuned LLM. Databricks blog, 2023. databricks-dolly-15k.

822 Junbo Cui, Bokai Xu, Chongyi Wang, Tianyu Yu, Weiyue Sun, Yingjing Xu, et al. MiniCPM-o 4.5:
823 Towards real-time full-duplex omni-modal interaction. *arXiv preprint arXiv:2604.27393*, 2026.

824 DeepSeek-AI, Daya Guo, et al. DeepSeek-R1: Incentivizing reasoning capability in LLMs via
825 reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.

826 Alexandre Défossez, Laurent Mazaré, Manu Orsini, Amélie Royer, Patrick Pérez, Hervé Jégou,
827 Edouard Grave, and Neil Zeghidour. Moshi: a speech-text foundation model for real-time dia-
828 logue. *arXiv preprint arXiv:2410.00037*, 2024.

829 Dujian Ding, Ankur Mallick, Chi Wang, Robert Sim, Subhabrata Mukherjee, Victor Rühle, Laks
830 V. S. Lakshmanan, and Ahmed Hassan Awadallah. Hybrid LLM: Cost-efficient and quality-
831 aware query routing. In *International Conference on Learning Representations (ICLR)*, 2024.
832 arXiv:2404.14618.

833 Fahim Faisal, Sharlina Keshava, Md Mahfuz ibn Alam, and Antonios Anastasopoulos. SD-QA:
834 Spoken dialectal question answering for the real world. In *Findings of the Association for Com-
835 putational Linguistics: EMNLP 2021*, 2021. arXiv:2109.12072.

836 Qingkai Fang, Shoutao Guo, and Yang Feng. BayLing-Duplex: Native full-duplex speech dialogue
837 with a single autoregressive LLM. *arXiv preprint arXiv:2606.14528*, 2026.

838 Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. Detecting hallucinations in large
839 language models using semantic entropy. *Nature*, 630:625–630, 2024.

840 Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song,
841 and Jacob Steinhardt. Measuring mathematical problem solving with the MATH dataset. In
842 *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks*, 2021.
843 arXiv:2103.03874.

844 Qitian Jason Hu, Jacob Bieker, Xiuyu Li, Nan Jiang, Benjamin Keigwin, Gaurav Ranganath, Kurt
845 Keutzer, and Shriyash Kaustubh Upadhyay. RouterBench: A benchmark for multi-LLM routing
846 systems. *arXiv preprint arXiv:2403.12031*, 2024.

847 Shengding Hu, Yifan Luo, Huadong Wang, Xingyi Cheng, Zhiyuan Liu, and Maosong Sun. Won’t
848 get fooled again: Answering questions with false premises. In *Proceedings of the 61st Annual
849 Meeting of the Association for Computational Linguistics (ACL)*, 2023. arXiv:2307.02394.

850 Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot,
851 Diego de las Casas, et al. Mistral 7B. *arXiv preprint arXiv:2310.06825*, 2023.

852 Mandar Joshi, Eunsol Choi, Daniel S. Weld, and Luke Zettlemoyer. TriviaQA: A large scale dis-
853 tantly supervised challenge dataset for reading comprehension. In *Proceedings of the 55th Annual*
854 *Meeting of the Association for Computational Linguistics (ACL)*, 2017.

855 Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez,
856 Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, et al. Language mod-
857 els (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.

858 Daniel Kahneman. *Thinking, Fast and Slow*. Farrar, Straus and Giroux, 2011.

859 Jannik Kossen, Jiatong Han, Muhammed Razzak, Lisa Schut, Shreshth Malik, and Yarin Gal. Se-
860 mantic entropy probes: Robust and cheap hallucination detection in LLMs. *arXiv preprint*
861 *arXiv:2406.15927*, 2024.

862 Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. Semantic uncertainty: Linguistic invariances for
863 uncertainty estimation in natural language generation. In *International Conference on Learning*
864 *Representations (ICLR)*, 2023. arXiv:2302.09664.

865 Yaniv Leviathan, Matan Kalman, and Yossi Matias. Fast inference from transformers via speculative
866 decoding. In *International Conference on Machine Learning (ICML)*, 2023. arXiv:2211.17192.

867 Chia-Hsuan Li, Szu-Lin Wu, Chi-Liang Liu, and Hung-yi Lee. Spoken SQuAD: A study of miti-
868 gating the impact of speech recognition errors on listening comprehension. In *Interspeech*, 2018.
869 arXiv:1804.00320.

870 Hao Li, Yiqun Zhang, Zhaoyan Guo, Chenxu Wang, Shengji Tang, Qiaosheng Zhang, Yang Chen,
871 Biqing Qi, Peng Ye, Lei Bai, Zhen Wang, and Shuyue Hu. LLMRouterBench: A massive bench-
872 mark and unified framework for LLM routing. *arXiv preprint arXiv:2601.07206*, 2026.

873 Borui Liao, Yulong Xu, Jiao Ou, Kaiyuan Yang, Weihua Jian, Pengfei Wan, and Di Zhang. Flex-
874 Duo: A pluggable system for enabling full-duplex capabilities in speech dialogue systems. *arXiv*
875 *preprint arXiv:2502.13472*, 2025.

876 Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan
877 Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s verify step by step. In *International*
878 *Conference on Learning Representations (ICLR)*, 2024. arXiv:2305.20050; source of the MATH-
879 500 subset.

880 Stephanie Lin, Jacob Hilton, and Owain Evans. Teaching models to express their uncertainty in
881 words. *Transactions on Machine Learning Research*, 2022. arXiv:2205.14334.

882 Mingbo Ma, Liang Huang, Hao Xiong, Renjie Zheng, Kaibo Liu, Baigong Zheng, Chuanqiang
883 Zhang, Zhongjun He, Hairong Liu, Xing Li, Hua Wu, and Haifeng Wang. STACL: Simultaneous
884 translation with implicit anticipation and controllable latency using prefix-to-prefix framework. In
885 *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*,
886 2019.

887 David Madras, Toniann Pitassi, and Richard Zemel. Predict responsibly: Improving fairness and
888 accuracy by learning to defer. In *Advances in Neural Information Processing Systems (NeurIPS)*,
889 2018.

890 Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi.
891 When not to trust language models: Investigating effectiveness of parametric and non-parametric
892 memories. In *Proceedings of the 61st Annual Meeting of the Association for Computational*
893 *Linguistics (ACL)*, 2023. PopQA; arXiv:2212.10511.

894 Samuel Marks and Max Tegmark. The geometry of truth: Emergent linear structure in large language
895 model representations of true/false datasets. In *Conference on Language Modeling (COLM)*, 2024.
896 arXiv:2310.06824.

897 Tu Anh Nguyen, Eugene Kharitonov, Jade Copet, Yossi Adi, Wei-Ning Hsu, Ali Elkahky, Paden
898 Tomasello, Robin Algayres, Benoît Sagot, Abdelrahman Mohamed, and Emmanuel Dupoux. Gen-
899 erative spoken dialogue language modeling. *Transactions of the Association for Computational*
900 *Linguistics*, 2023. arXiv:2203.16502.

901 Isaac Ong, Amjad Almahairi, Vincent Wu, Wei-Lin Chiang, Tianhao Wu, Joseph E. Gonzalez,
902 M. Waleed Kadous, and Ion Stoica. RouteLLM: Learning to route LLMs with preference data.
903 *arXiv preprint arXiv:2406.18665*, 2024.

904 Hadas Orgad, Michael Toker, Zorik Gekhman, Roi Reichart, Idan Szpektor, and Yonatan Belinkov.
905 LLMs know more than they show: On the intrinsic representation of LLM hallucinations. In
906 *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2410.02707. TODO
907 verify author list.

908 Pablo Riera, Pablo Brusco, Cristina Kuo, Marcelo Sancinetti, and S. R. K. Branavan. Synchroniza-
909 tion and turn-taking in full-duplex speech dialogue models. *arXiv preprint arXiv:2605.20356*,
910 2026.

911 Tal Schuster, Adam Fisch, Jai Gupta, Mostafa Dehghani, Dara Bahri, Vinh Q. Tran, Yi Tay, and
912 Donald Metzler. Confident adaptive language modeling. In *Advances in Neural Information*
913 *Processing Systems (NeurIPS)*, 2022. arXiv:2207.07061.

914 Tanya Stivers, N. J. Enfield, Penelope Brown, Christina Englert, Makoto Hayashi, Trine Heinemann,
915 Gertie Hoymann, Federico Rossano, Jan Peter de Ruiter, Kyung-Eun Yoon, and Stephen C. Levin-
916 son. Universals and cultural variation in turn-taking in conversation. *Proceedings of the National*
917 *Academy of Sciences*, 106(26):10587–10592, 2009.

918 Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy
919 Liang, and Tatsunori B. Hashimoto. Stanford alpaca: An instruction-following LLaMA model.
920 GitHub repository, 2023. zh split: shibing624/alpaca-zh.

921 Team GLM, Aohan Zeng, et al. ChatGLM: A family of large language models from GLM-130B to
922 GLM-4 all tools. *arXiv preprint arXiv:2406.12793*, 2024.

923 Tu Vu, Mohit Iyyer, Xuezhi Wang, Noah Constant, Jerry Wei, Jason Wei, Chris Tar, Yun-Hsuan
924 Sung, Denny Zhou, Quoc Le, and Thang Luong. Freshllms: Refreshing large language models
925 with search engine augmentation. *arXiv preprint arXiv:2310.03214*, 2023.

926 Xiong Wang, Yangze Li, Chaoyou Fu, Lei Xie, Ke Li, Xing Sun, and Long Ma. Freeze-Omni:
927 A smart and low latency speech-to-speech dialogue model with frozen LLM. *arXiv preprint*
928 *arXiv:2411.00774*, 2024a.

929 Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, et al. MMLU-Pro: A more robust and
930 challenging multi-task language understanding benchmark. In *Advances in Neural Information*
931 *Processing Systems (NeurIPS), Datasets and Benchmarks*, 2024b. arXiv:2406.01574.

932 Jason Wei, Nguyen Karina, Hyung Won Chung, Yunxin Joy Jiao, Spencer Papay, Amelia Glaese,
933 John Schulman, and William Fedus. Measuring short-form factuality in large language models.
934 *arXiv preprint arXiv:2411.04368*, 2024. SimpleQA.

935 Donghang Wu, Haoyang Zhang, Chen Chen, Tianyu Zhang, Fei Tian, Xuerui Yang, Gang Yu, Hexin
936 Liu, Nana Hou, Yuchen Hu, and Eng Siong Chng. Chronological thinking in full-duplex spoken
937 dialogue language models. In *Proceedings of the 27th Annual Meeting of the Special Interest*
938 *Group on Discourse and Dialogue (SIGDIAL)*, 2026. arXiv:2510.05150.

939 Jin Xu, Zhifang Guo, Jinzheng He, Hangrui Hu, Ting He, Shuai Bai, Keqin Chen, et al. Qwen2.5-
940 Omni technical report. *arXiv preprint arXiv:2503.20215*, 2025.

941 Leyang Xue, Yao Fu, Luo Mai, and Mahesh K. Marina. HybridServe: Efficient serving of large AI
942 models with confidence-based cascade routing. *arXiv preprint arXiv:2505.12566*, 2025.

943 An Yang et al. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.

944 An Yang et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.

945 Yuan Yao, Tianyu Yu, Ao Zhang, Chongyi Wang, Junbo Cui, Hongji Zhu, et al. MiniCPM-V: A
946 GPT-4V level MLLM on your phone. *arXiv preprint arXiv:2408.01800*, 2024.

947 Aohan Zeng, Zhengxiao Du, Mingdao Liu, Kedong Wang, Shengmin Jiang, Lei Zhao, Yuxiao Dong,
948 and Jie Tang. GLM-4-Voice: Towards intelligent and human-like end-to-end spoken chatbot.
949 *arXiv preprint arXiv:2412.02612*, 2024.

950 Dengjia Zhang, Alexander Martin, William Jurayj, Kenton Murray, Benjamin Van Durme, and Reno
951 Kriz. Unified multimodal uncertain inference. *arXiv preprint arXiv:2604.08701*, 2026.

952 Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang,
953 Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica.
954 Judging LLM-as-a-judge with MT-Bench and chatbot arena. In *Advances in Neural Information*
955 *Processing Systems (NeurIPS), Datasets and Benchmarks*, 2023. arXiv:2306.05685.

TODO / Roadmap (working section — delete before submission)

[W] Must-write — ALL EXECUTED 2026-08-25 (peer-draft session)

- **[TODO: W1 — BIGGEST GAP:** the external-benchmark transferability section does not exist in the paper. live.tex has ZERO mention of striviaqa / swabq / sdqa / sllama / sreason / valpaca; none of the 12 dualview/pareto family figures are included. This was the 2026-08-12 meeting deliverable. Sources: RESULTS §8x (bench infra + latency shapes), §8z + 8z-live (v3 gains: striviaqa balanced .764→.800, sreason cross-lingual, sllama selective@50% .948), §8ab (framing rules). Anchors per figure: official MiniCPM-o 4.5 number (floor; 75.5 striviaqa / 70.2 swabq), GPT-5.5-on-transcript (ceiling), random escalation at matched rate (control). Framing rules from the Jisen pass (8ab): (a) cite that the gate dominates the simulated-P50 random curve in BOTH axes on striviaqa/sllama; (b) the P50 left-fold is a verified median-of-a-mixture mechanism — use sllama_latency_decomp, report mean alongside median; (c) frame router quality via AUC + matched-rate precision (v3 hits 95% of the base-fail/rate cap on sllama), never raw precision; latency-shape note: non-monotonic per-arm P50 on easy pools, tail truncation on Reasoning QA (P90 13.3→10.9 s)]
- **[TODO: W2 — 8ad noise audit: THREE MANDATORY REWRITES of §8,** still absent from live.tex (verified 2026-08-25: Table 8 still carries the v1 numbers .400/.446/.529/.633). (a) re-read every arm number off live_v3_figures.py output: heard .383/.483/.554/.596 at 0/16/35/61%, channel cost −.175, P50 2.02/2.61/3.53/4.44 s, P99 32.7 s at balanced. (b) sllama headline: “MATCHES the expert at half the cost” (.976 vs .968 McNemar $p=1.00$); what is significant is the probe’s split ($z=6.46$) and the hard-half lift .696→.888 ($p<.0001$); .948-vs-.928 is $p=.125$. (c) report the replication-noise floor (same-query flip 2.3–18.8%, paired SE .009–.028), stop narrating sub-3-point live deltas (all 18 v2→v3 arm deltas n.s.; v3 evidence is the offline AUC). Add noise_audit as a methods figure]
- **[TODO: W3 — second duplex family (8ac, EXECUTED 2026-08-19, not in any section):** NemotronLabs-VoiceChat-11B (hybrid Mamba), the pre-registered prediction test — mid-band layer structure replicates (L30–34 of 56), frozen-methodology transfer AUC .70–.79 with 600-query calib; sreason boundary (model is English-only). One paragraph in duplex.tex or discussion: this is the generality evidence for “mid-network read” beyond MiniCPM]
- **[TODO: W4 — cloud-ASR uplink (8ae, not in any section):** on 147 escalated frozen ids, gpt-transcribe → gpt-5.5 scores .694 vs .585 deployed self-transcript (McNemar $p=.007$), recovering 38% of the gap to the .871 gold ceiling; SUPERSEDES the “Whisper ears +4pp only” write-off (8d used whisper-large-v3). Write as diagnostic finding with both caveats (ASR latency on critical path unmeasured; external-pool survival unmeasured). hard-math recovers nothing — spoken LaTeX is an input-side limit, state it separately where input-side limits are stated]
- **[TODO: W5 — small syncs:** (a) deployment-artifact sentence: label-free quantile correction (gate_v3_thresholds_corrected.json) removes 11.3% of expert calls at equal accuracy on the frozen pool; externals already on-budget. (b) decide whether the demo system (8aj–8am: TTS head, v4 real-time gate, web-search expert, backchannel-aware barge-in) gets a short system/demo paragraph or an appendix; it is currently only in RESULTS. (c) FreshQA/v4 paragraph already in live.tex — verify numbers match RESULTS 8ak final (.877 frozen-test, fire .80/.93/1.0)]

[D] Decisions — ALL RATIFIED 2026-08-25 (user approved all three)

- **[TODO: D1 — speakable-subset headline swap (8q, user to ratify):** fairness audit removes 22/240 ids (21 LaTeX + 1 broken wav; criteria decidable from query text + wav bytes alone) ⇒ aggressive heard-acc .633→.688, channel gap .133→.083 (still

significant), gold ceiling unmoved. If ratified: dual-view + Pareto redrawn on the fair subset (fair_* figures already built), full-pool to appendix, criteria stated as pre-registered filters, update the 8m 64%-corrupted sentence in live.tex. NOTE: interacts with W2 — do the v1→v3 number swap and this in one pass]

- [TODO: D2 — DONE (2026-08-25, RESULTS §8an, scripts/07_bootstrap_oracle.py): point estimates reproduced (+.0541/+0.0422/+0.0119); paired bootstrap $B=10^4$ gives 95% CI $[-.003, +.027]$, $p=.13$ — NOT significant, as the 7-29 entry suspected. system.tex:21 softened to “statistically on par with the pool oracle in distribution”. No other untested-delta claim of this class remains]
- [TODO: D3 — double-blind exposure: the demo URL (rhe9527-gate-demo-web.modal.run) contains the author username; author block on the title page must go for submission (camera-ready restores it). Decide: anonymized demo link (new modal workspace / redacted) vs. “demo available upon publication”]

1021 [M] Mechanical

- [TODO: M1 — DONE 2026-08-25: neurips_2026.sty submission mode, author block anonymized, compiles under tectonic. Remaining: the 8-page main-text compression (currently far over)]
- [TODO: M2 — figure hygiene: (a) regenerate concept_probe_vs_ppl with English labels; (b) ROC figure to include $p(\text{True})$ curves; (c) verify pareto_latency rebuild (2026-08-20) actually contains the random curve (8ab-d); (d) vintage audit — any figure included in the paper with an artifact predating 2026-08-18 must be checked against v3 traces (currently included: concept_probe_vs_ppl, fork_pareto, layer_sweep, live_dualview, overlap, pareto_latency, receipt_compare, roc, timeline_live, tradeoff, tradeoff_midlayer, tradeoff_pttrue — the July-vintage ones are suspect)]
- [TODO: M3 — bib: verify Li et al. [2026] (arXiv:2601.07206) and the MiniCPM-o 4.5 model-card entry]
- [TODO: M4 — surface existing bootstrap CIs in tables beyond Table 2]
- [TODO: M5 — appendix assembly: full per-pool tables, judge prompts, $p(\text{True})$ prompts, reproducibility statement (seeds, pins, spend)]

1038 [F] Future work — fold into discussion, then delete

- [TODO: one paragraph covering: (a) 10 remaining SD-QA dialect splits (mechanical robustness table, ~\$100); (b) true duplex integration — ls-head + L22 probe validity under the duplex loop (the probe is calibrated on simplex streaming states); (c) Nemotron remaining sub-items (live 4-arm port, calib expansion, trained-checkpoint ablation); (d) audio-direct-to-expert live sweep (the one lever a single sweep can resolve; gold/heard gap $-.175$ is an order above the noise floor); (e) trained-gate upper-bound baseline (the only legitimate “real training”: a contrast, not a method change); (f) LLMRouterBench protocol comparison; (g) $p(\text{True})$ prompt-sensitivity sweep; (h) injection payload knee (answer + one sentence justification vs. math relay compliance, \$5); (i) closed-source behavioral arm (GPT audio / Gemini Live on the 600 ways); (j) camera-ready: fresh test split + pool expansion to 500–1000/pool]